

海光BW1000、英伟达H100 - 单节点MiniMax-M2.5模型整体测试比对报告

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*测试人员：九章云极

1. 测试背景

公司需要在多个候选开源大模型中选型，部署基于vLLM或SGLang的推理服务。并需要在满足各项模型服务指标的情况下，选定芯片集采厂商。

2. 测试目标

本测试主要评估不同芯片在单机环境下运行大模型推理的能力，为后续集群采购和生产部署提供决策依据。

- 硬件摸底**：确认各芯片型号实际规格（算力、显存、带宽）与标称值的一致性
- 功能验证**：各模型在各芯片环境上的推理正确性和算子兼容性
- 性能基准**：吞吐量、延迟、显存效率等关键指标
- 单机极限**：8卡 Tensor Parallel 的性能上限和资源利用率
- 稳定性验证**：长时间运行下的可靠性
- K8S 容器化验证**：单节点 K8S 环境下GPU/DCU/XPU 调度、资源管理和服务编排能力

3.1 硬件规格

组件 \ 规格	英伟达	海光	状态
节点数量	1 台	1 台	确认
芯片型号	H100	BW1000	确认
芯片数量	8 张	8 张	确认
单卡算力 FP16/BF16	1979 TFLOPS （官方理论值）	待确认	⚠ 待确认
单卡算力 FP32	67 TFLOPS （官方理论值）	待确认	⚠ 待确认
单卡算力 FP64	34 TFLOPS （官方理论值）	待确认	⚠ 待确认
单卡显存	80GB	64GB	确认
显存类型	HBM3	待确认	⚠ 待确认
显存带宽	3.35 TB/s	待确认	⚠ 待确认
单卡功耗	700 W	200 W	确认
卡间互联	NVLink 4.0	HSM	确认
CPU	Intel(R) Xeon(R) Platinum 8468 (192核)	Hygon C86 (128核)	确认
系统内存	2.0 TiB	503 GiB	确认
本地存储	894GB 系统盘 + 7TB*4 缓存盘 + 7TB 容器盘 + 25TB 扩展盘	437G系统盘 + 1.7TiB (G73M1T9R-C-GD308)	确认

3.2 软件栈

组件\版本	英伟达	海光	说明
操作系统	Ubuntu 22.04.5 LTS	Ubuntu 22.04.5 LTS	芯片所在物理机系统
显卡驱动	570.133.20/580.126.09	6.3.22-V1.2.0	驱动信息
Toolkit	release 12.9	DTK-26.04-beta-0130-ubuntu20.04	CUDA Toolkit版本
Docker	-	28.0.4	容器运行时
containerd	2.2.0	2.1.1	K8S 容器运行时（CRI）
Kubernetes	1.34.2	1.33	单节点 All-in-One 部署
Device Plugin	0.14.5	v2.4.0	K8S GPU 资源管理
多卡通信库	NCCL	DTK内置RCCL	多卡通信库

参数名称	NVIDIA_H100	Hygon_BW1000
model_name	MiniMax-M2.5	MiniMax-M2.5-W8A8
quantization_config	FP8	int-8
model_size	215G	215G
max_position_embeddings	196608	196608
temperature	1.0	N/A
top_k	40	N/A
top_p	0.95	N/A
transformers_version	4.46.1	4.57.6
vllm_version	0.20.0	0.15.1+das.opt1.alpha.dtk2604
python_version	3.12.3	3.10.12

3.4 推理框架主要启动参数

参数名称	NVIDIA_H100	Hygon_BW1000
Model Name	MiniMax-M2.5	MiniMax-M2.5-W8A8
Max Model Len	196608	196608
Max Num Seqs	64	64
Max Num Batched Tokens	8192	default
Block Size	default	default
Gpu Memory Utilization	0.85	0.9
Compilation Config	N/A	N/A
Dtype	default	bfloat16
Dp	1	1
Tp	8	8
Pp	1	1
Reasoning Parser	minimax_m2	minimax_m2
Tool Call Parser	minimax_m2	minimax_m2
Enable Auto Tool Choice	True	True
Enable Export Parallel	True	/

4.1 测试场景列表

序号	测试场景
场景一	vllm benchmark基准测试
场景二	单、多并发超长上下文请求
场景三	多并发长上下文极限验证
场景四	多I/O测试
场景五	模型精度测试
场景六	多轮对话测试
场景七	模型推理功能测试

4.2 模型部署问题汇总

N/A

4.3 模型推理测试问题汇总

N/A

以下是每个测试场景的详细结果报告

Hygon_BW1000平台，Minimax-M2.5模型benchmark测试脚本见本报告 《附录一》

测试场景一：vllm benchmark基准测试

测试目标：在相同请求数、基础长度上下文参数下，使用vllm bench serve工具对并发数逐级增加场景的性能基准验证.

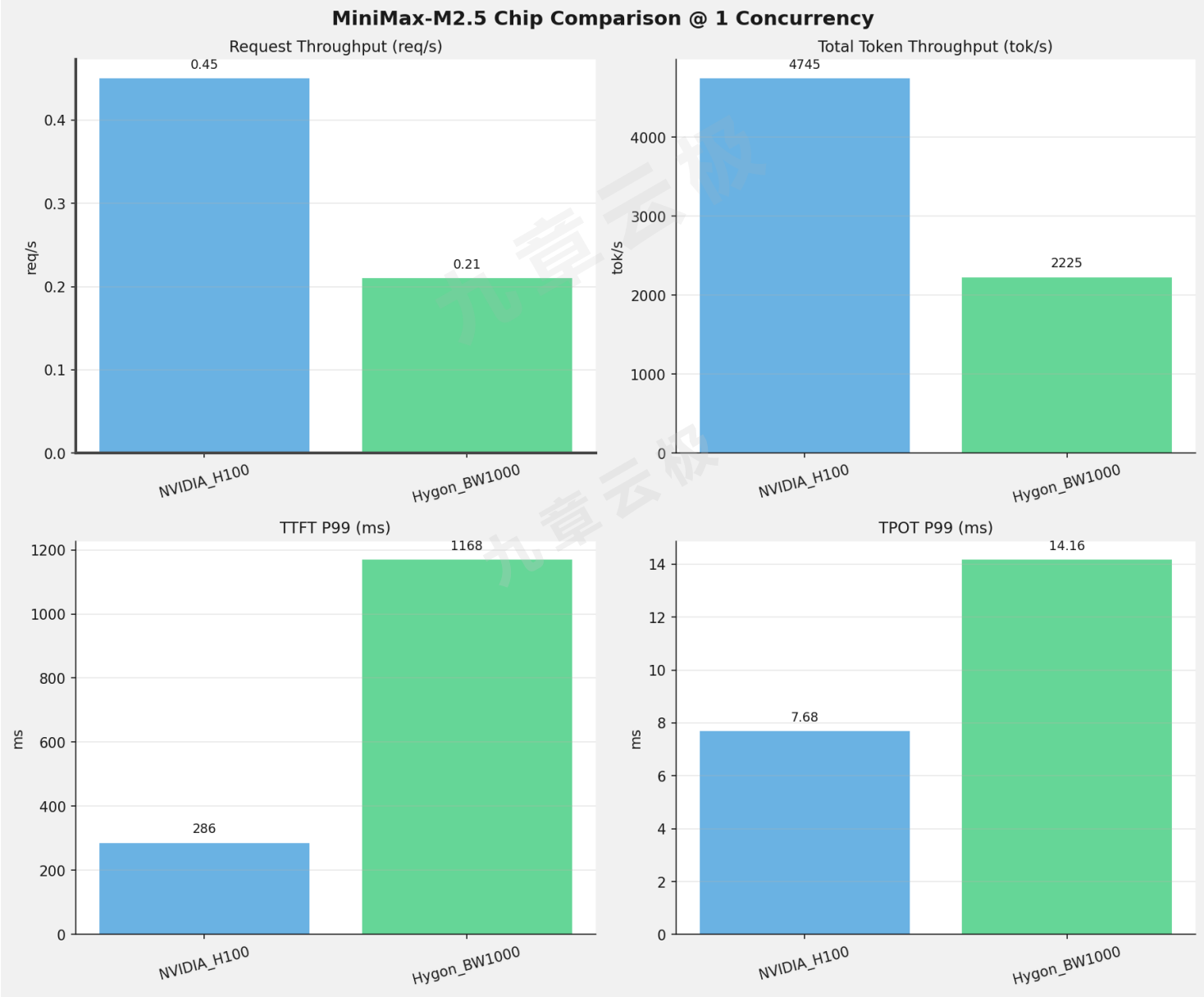
主要采集指标：

指标	单位	含义
TTFT	ms	Time To First Token，首 token 延迟
TPOT	ms/token	Time Per Output Token，每 token 生成时间
Throughput	tokens/s	系统总吞吐
QPS	requests/s	请求吞吐
P95/P99 Latency	ms	延迟分位数

项目	配置	备注
数据集	random	
并发数	1, 2, 4, 8, 10, 16, 32, 64, 80, 128	
总请求数	320	
请求输入上下文长度	10240 (10k)	
请求输出上下文长度	256 (0.25k)	
被测芯片	NVIDIA_H100, Hygon_BW1000	
被测模型	NVIDIA_H100 (MiniMax-M2.5); Hygon_BW1000 (MiniMax-M2.5-W8A8)	

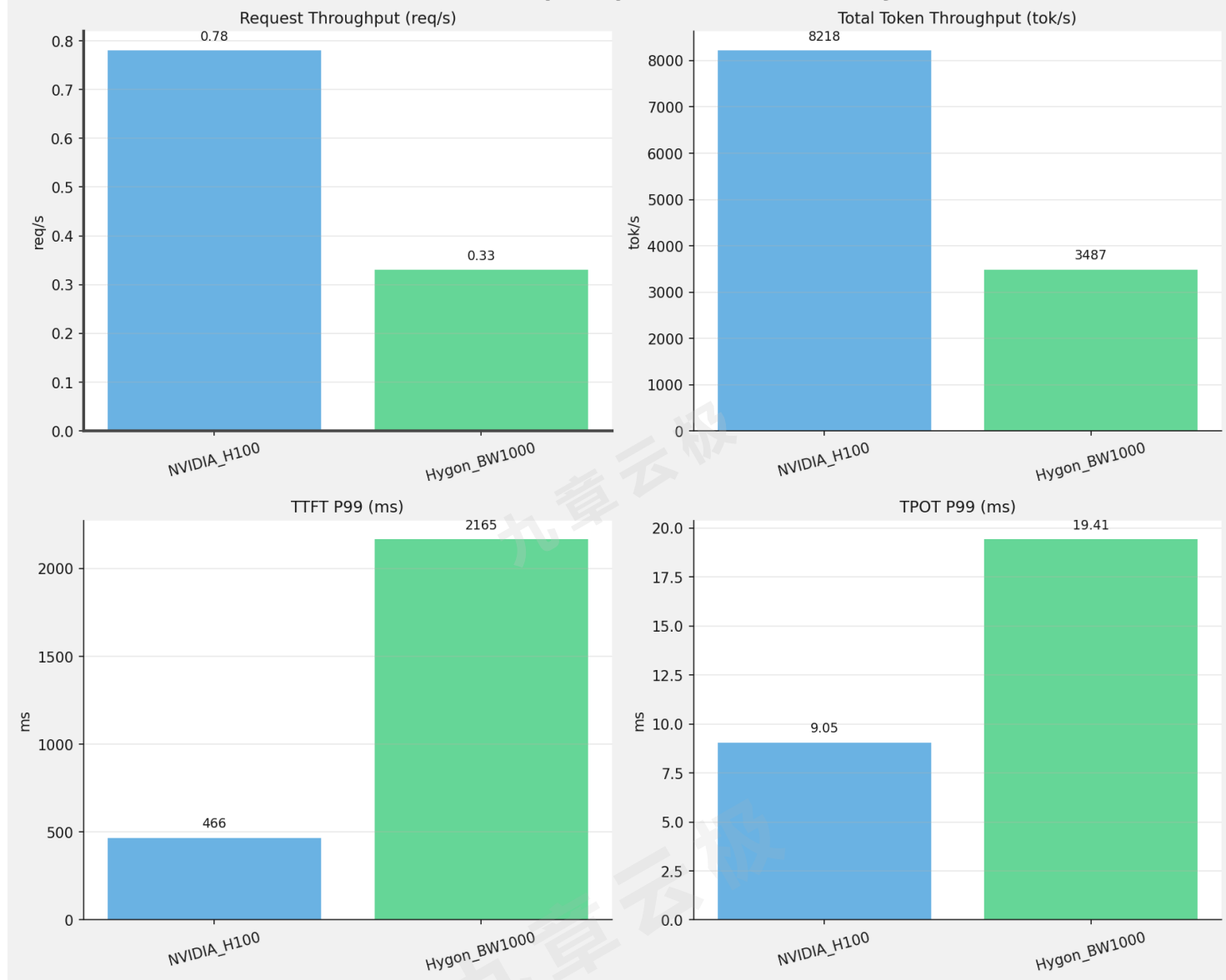
芯片性能对比柱状图

1并发



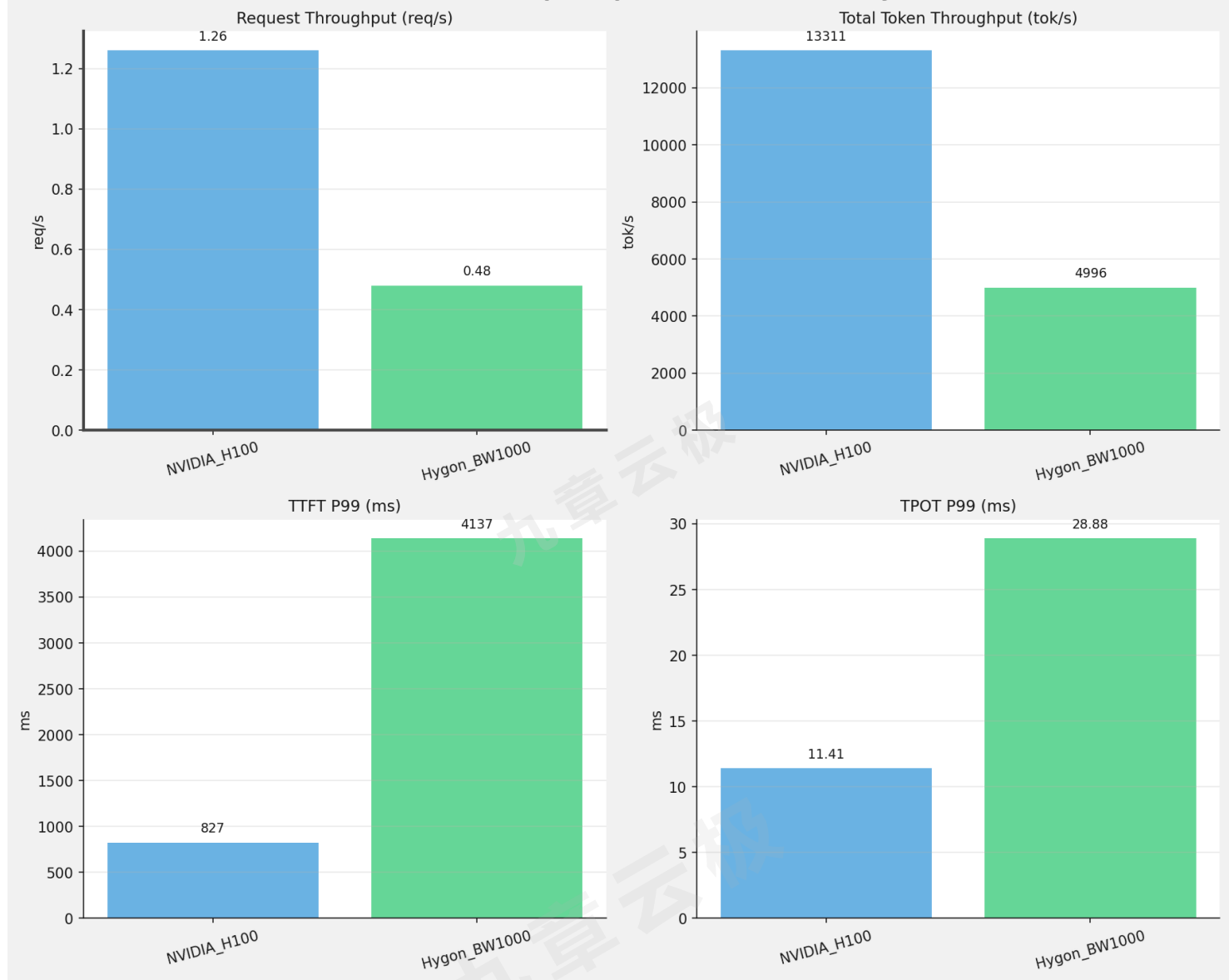
2并发

MiniMax-M2.5 Chip Comparison @ 2 Concurrency



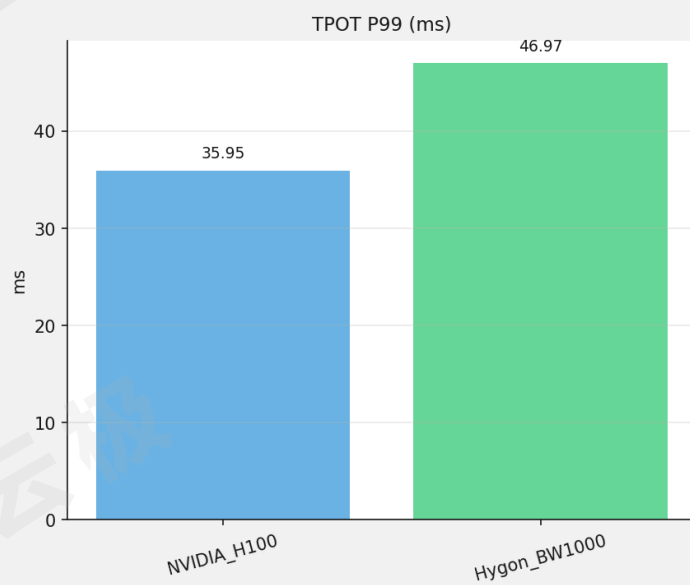
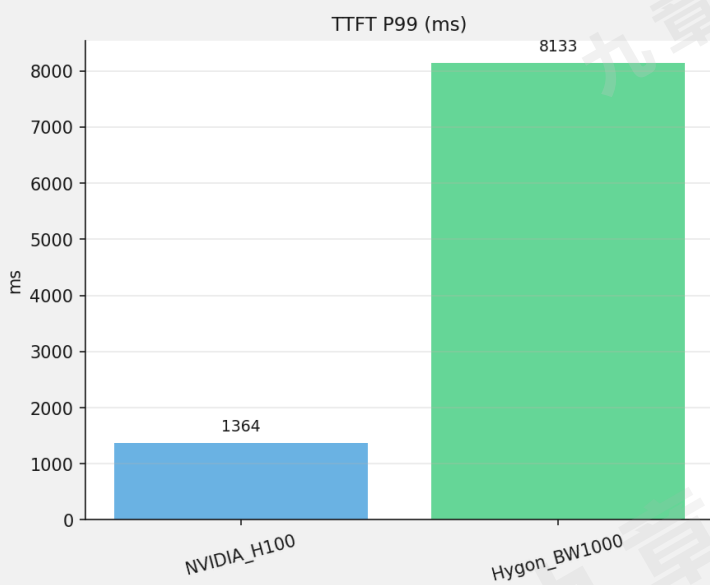
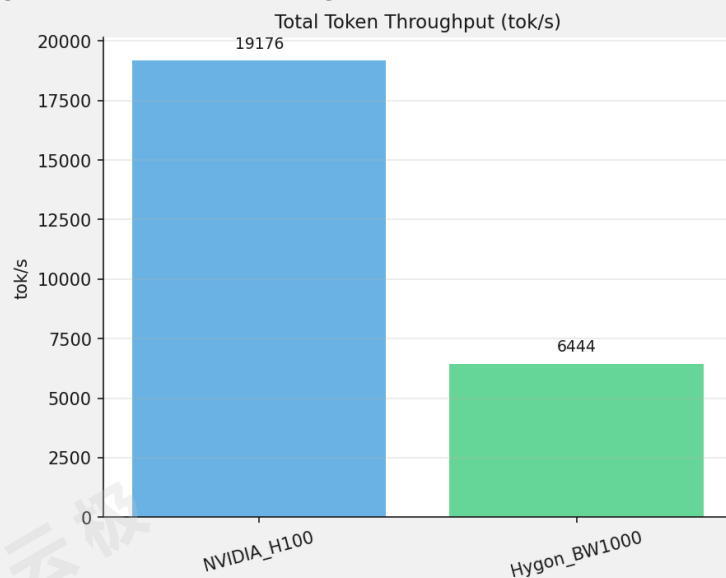
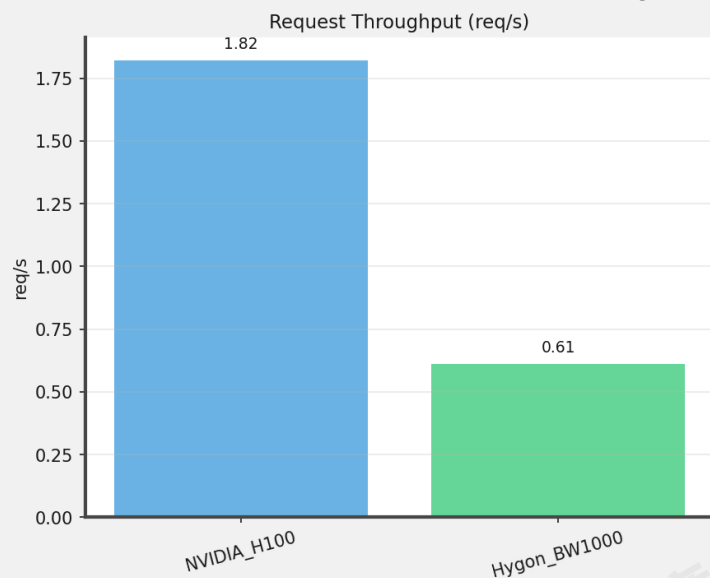
4并发

MiniMax-M2.5 Chip Comparison @ 4 Concurrency



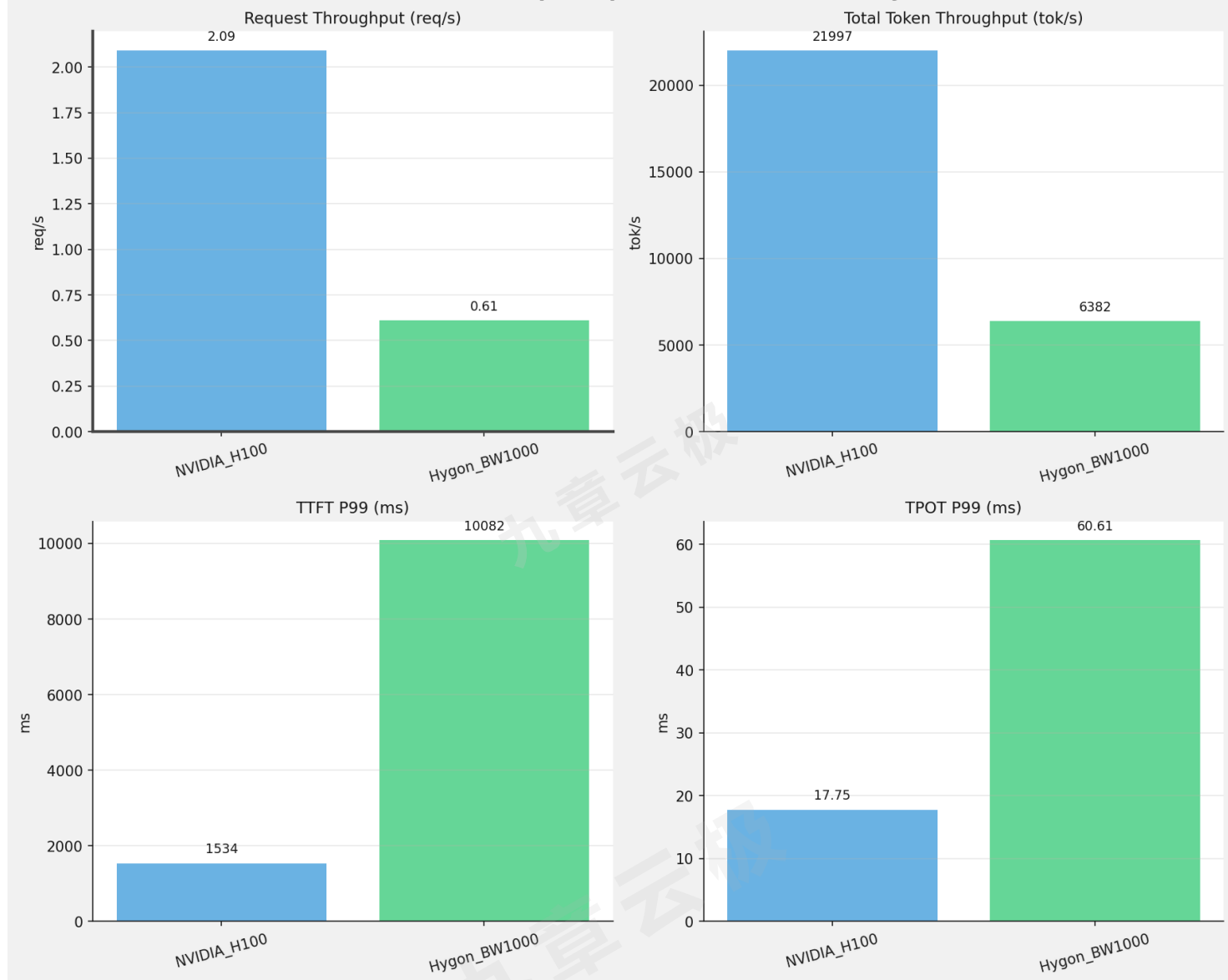
8并发

MiniMax-M2.5 Chip Comparison @ 8 Concurrency



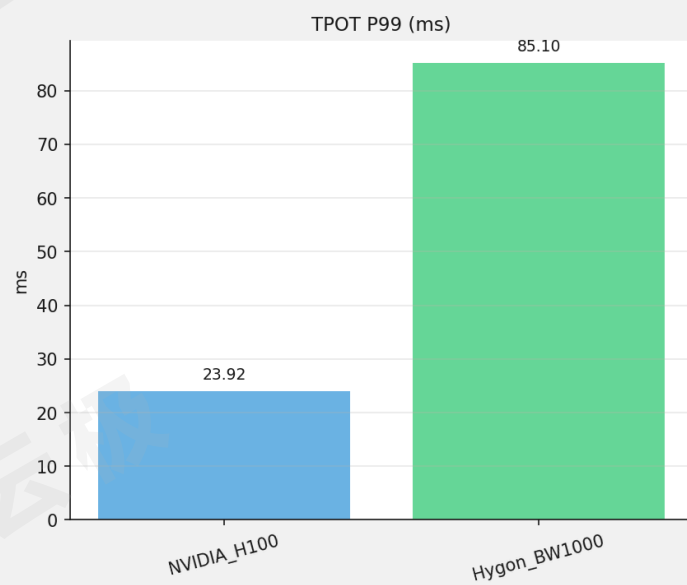
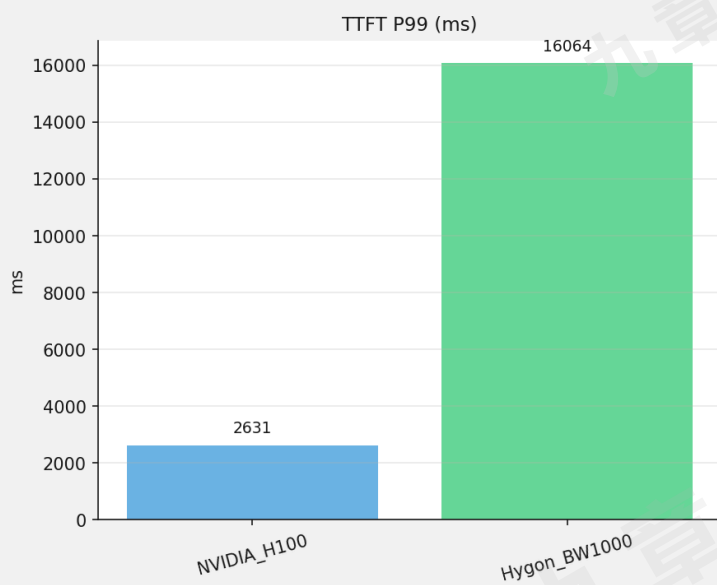
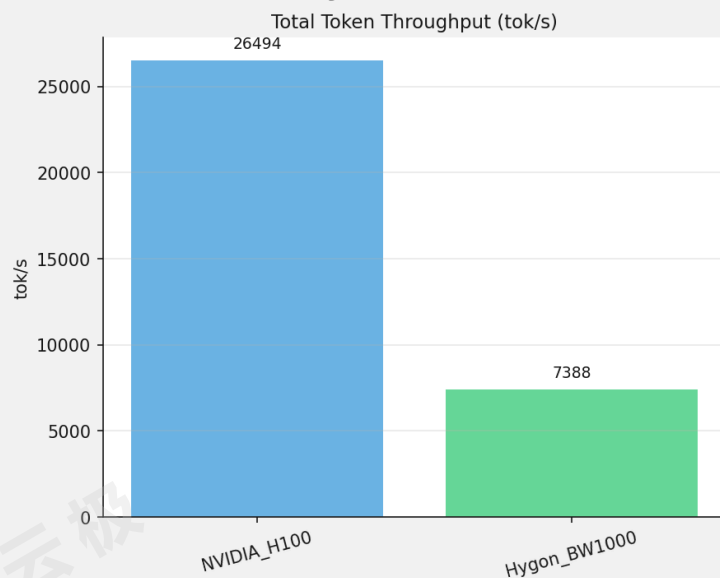
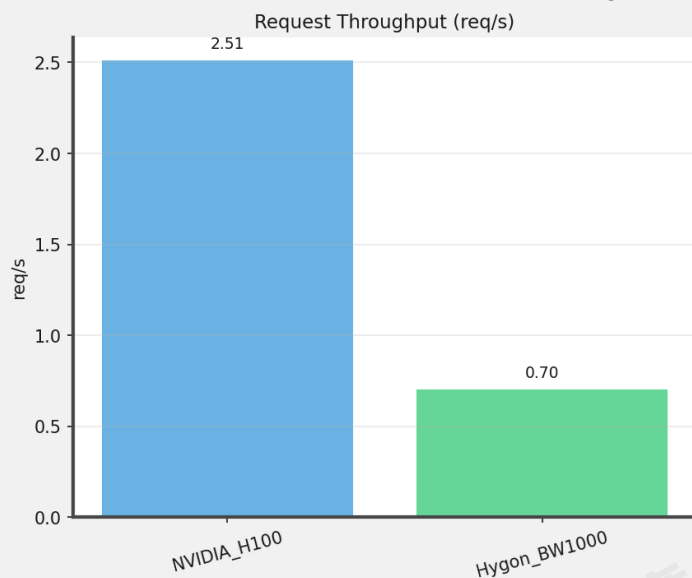
10并发

MiniMax-M2.5 Chip Comparison @ 10 Concurrency



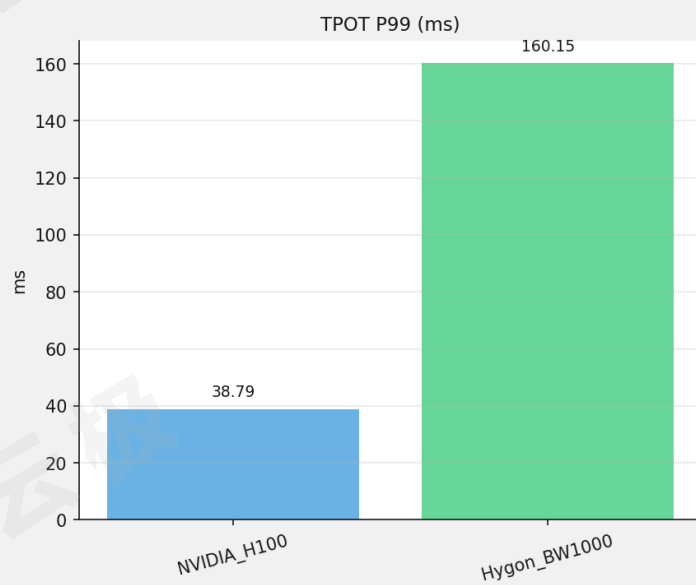
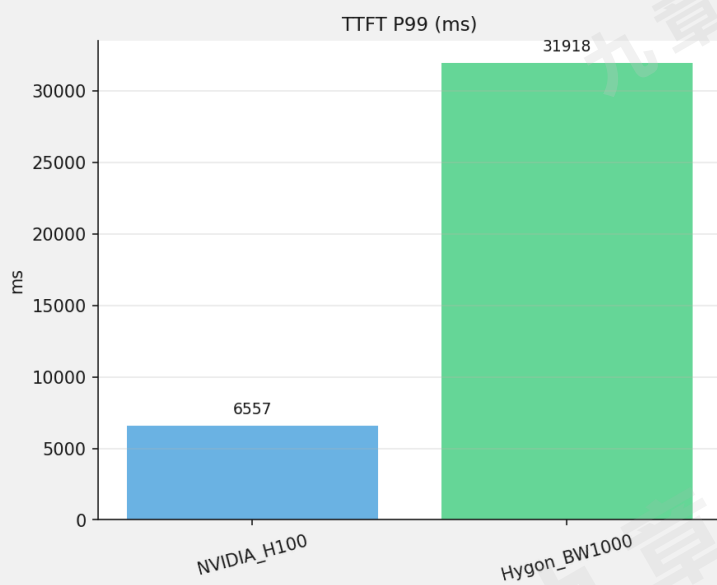
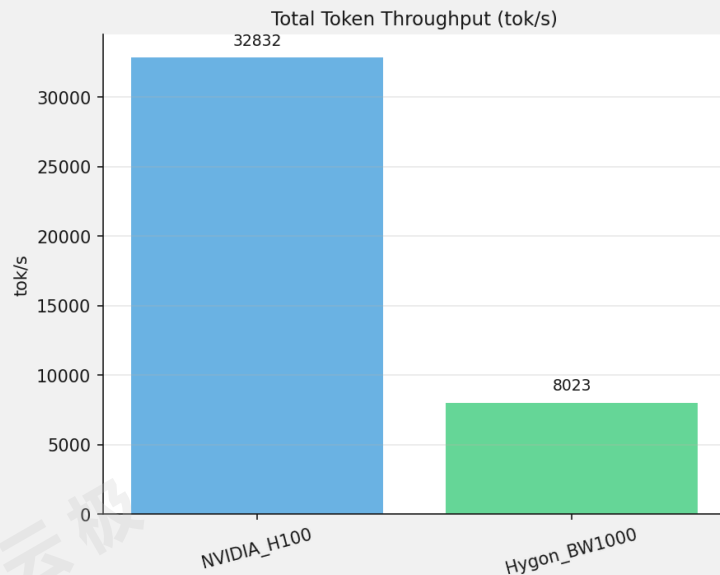
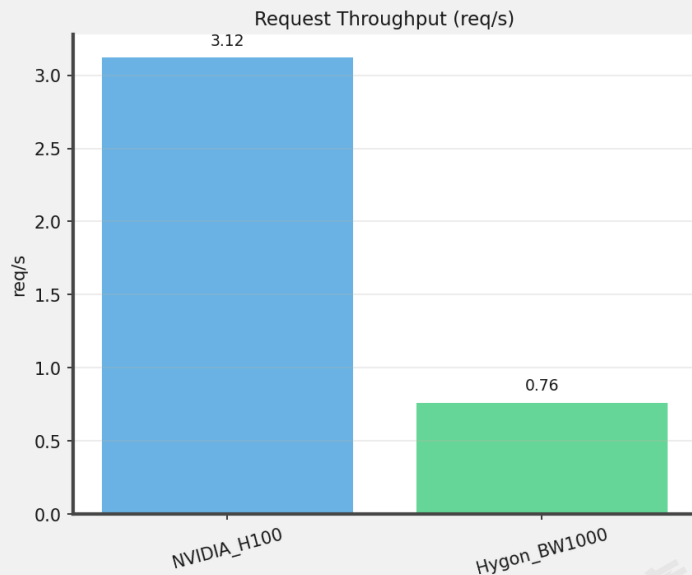
16并发

MiniMax-M2.5 Chip Comparison @ 16 Concurrency



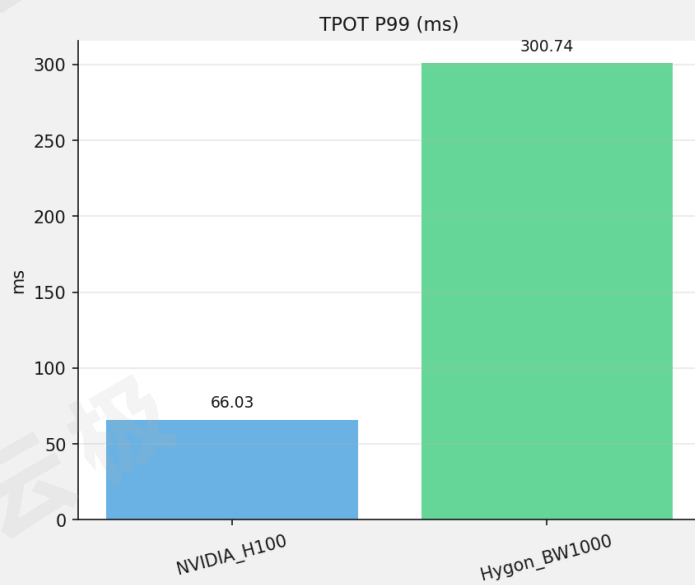
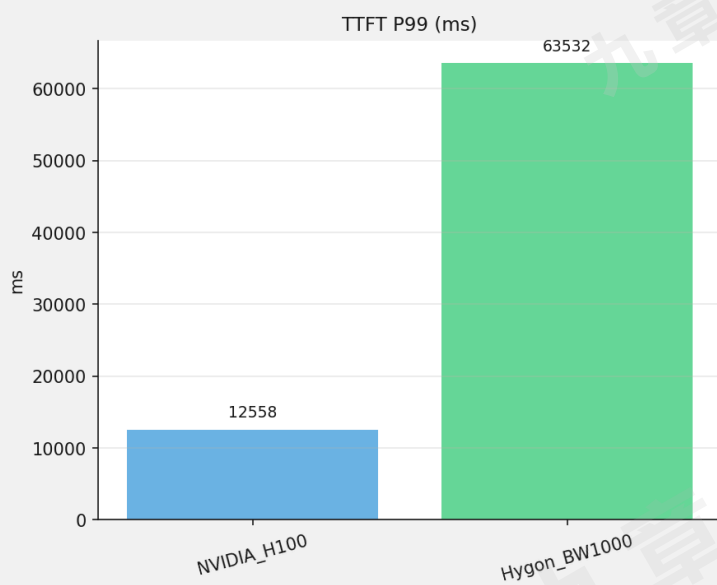
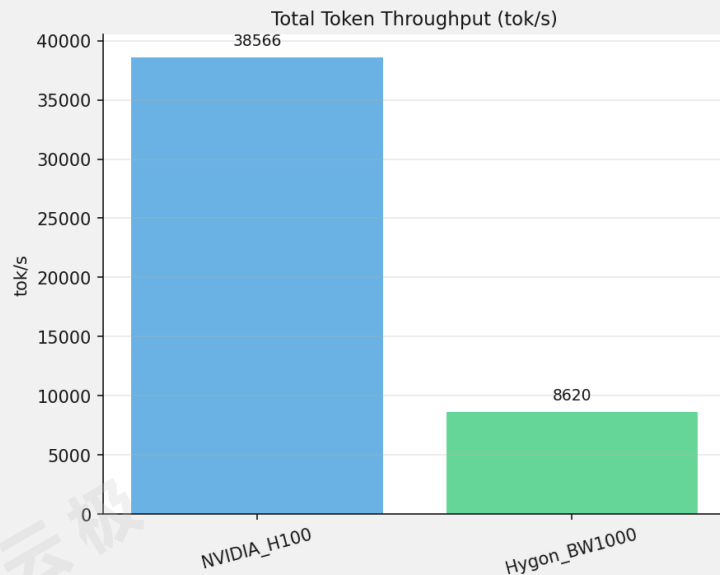
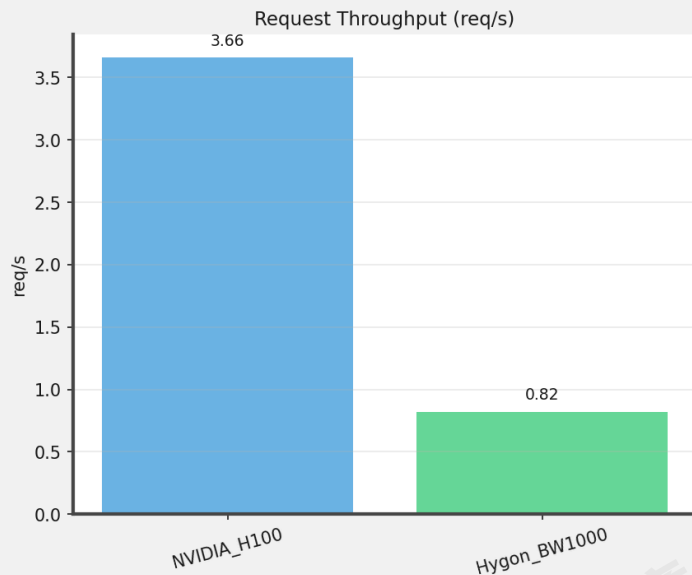
32并发

MiniMax-M2.5 Chip Comparison @ 32 Concurrency



64并发

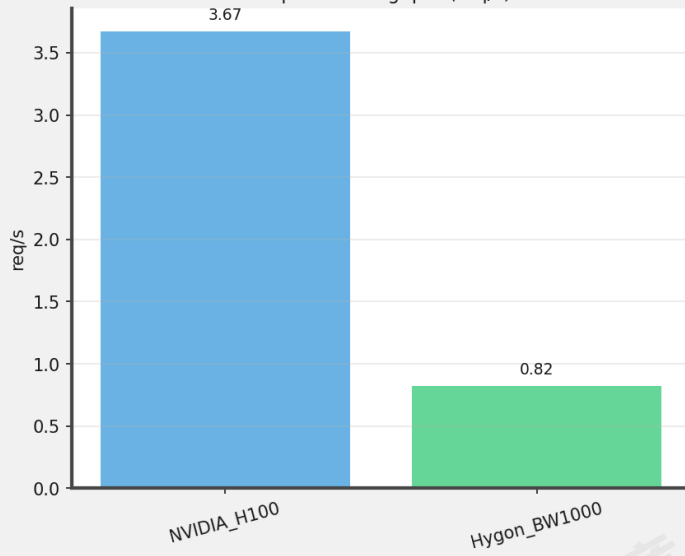
MiniMax-M2.5 Chip Comparison @ 64 Concurrency



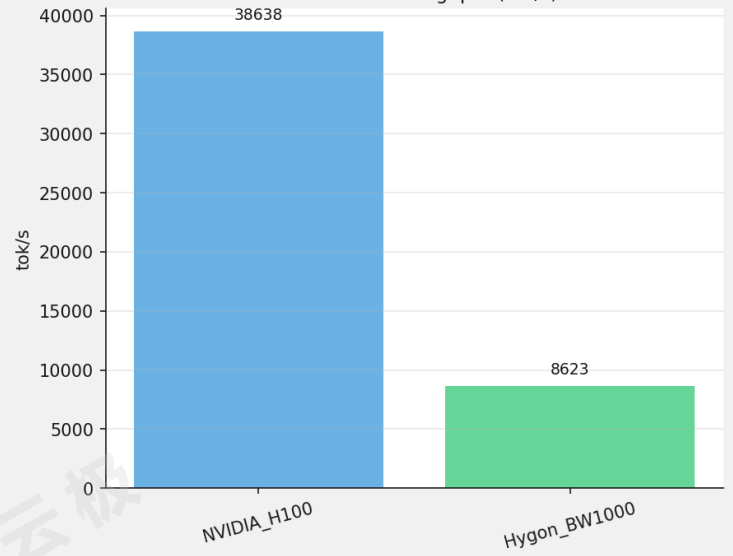
80并发

MiniMax-M2.5 Chip Comparison @ 80 Concurrency

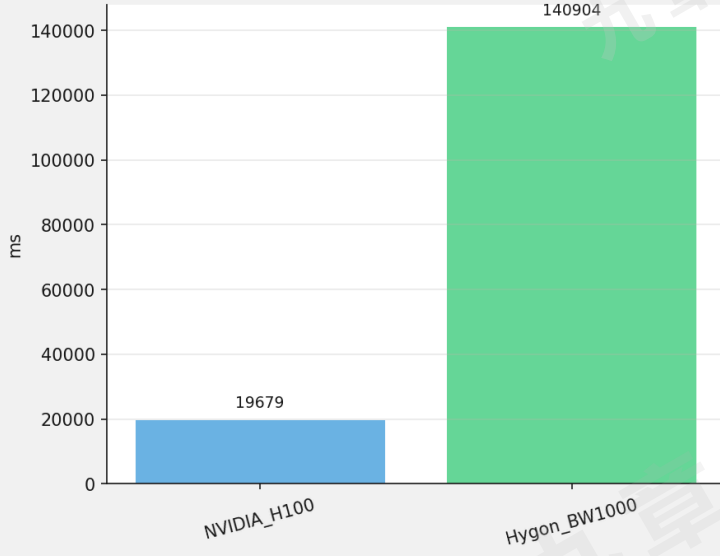
Request Throughput (req/s)



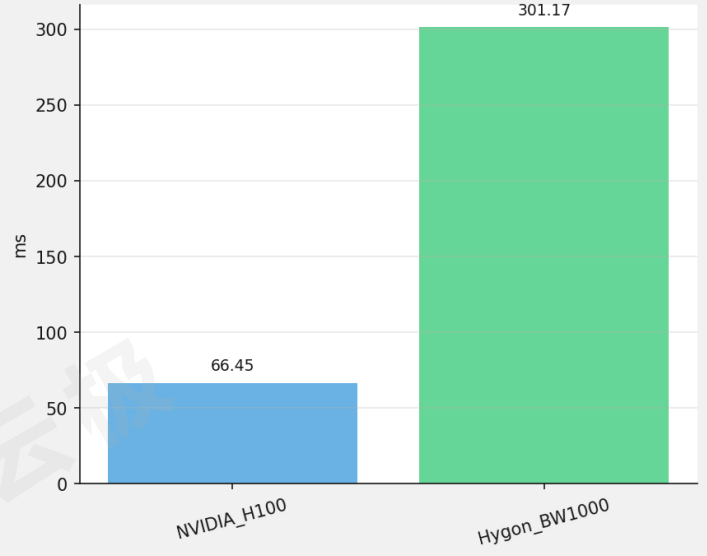
Total Token Throughput (tok/s)



TTFT P99 (ms)

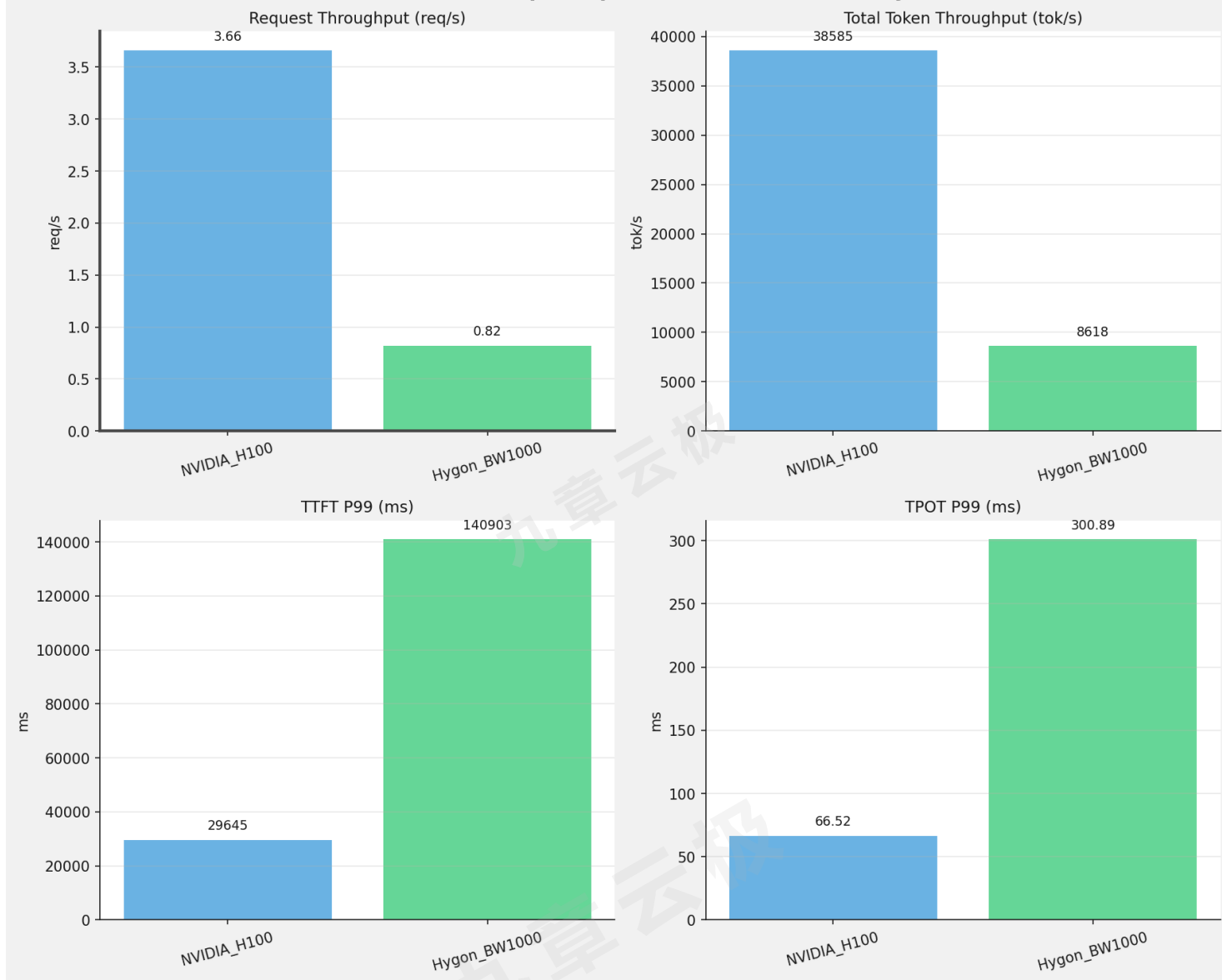


TPOT P99 (ms)



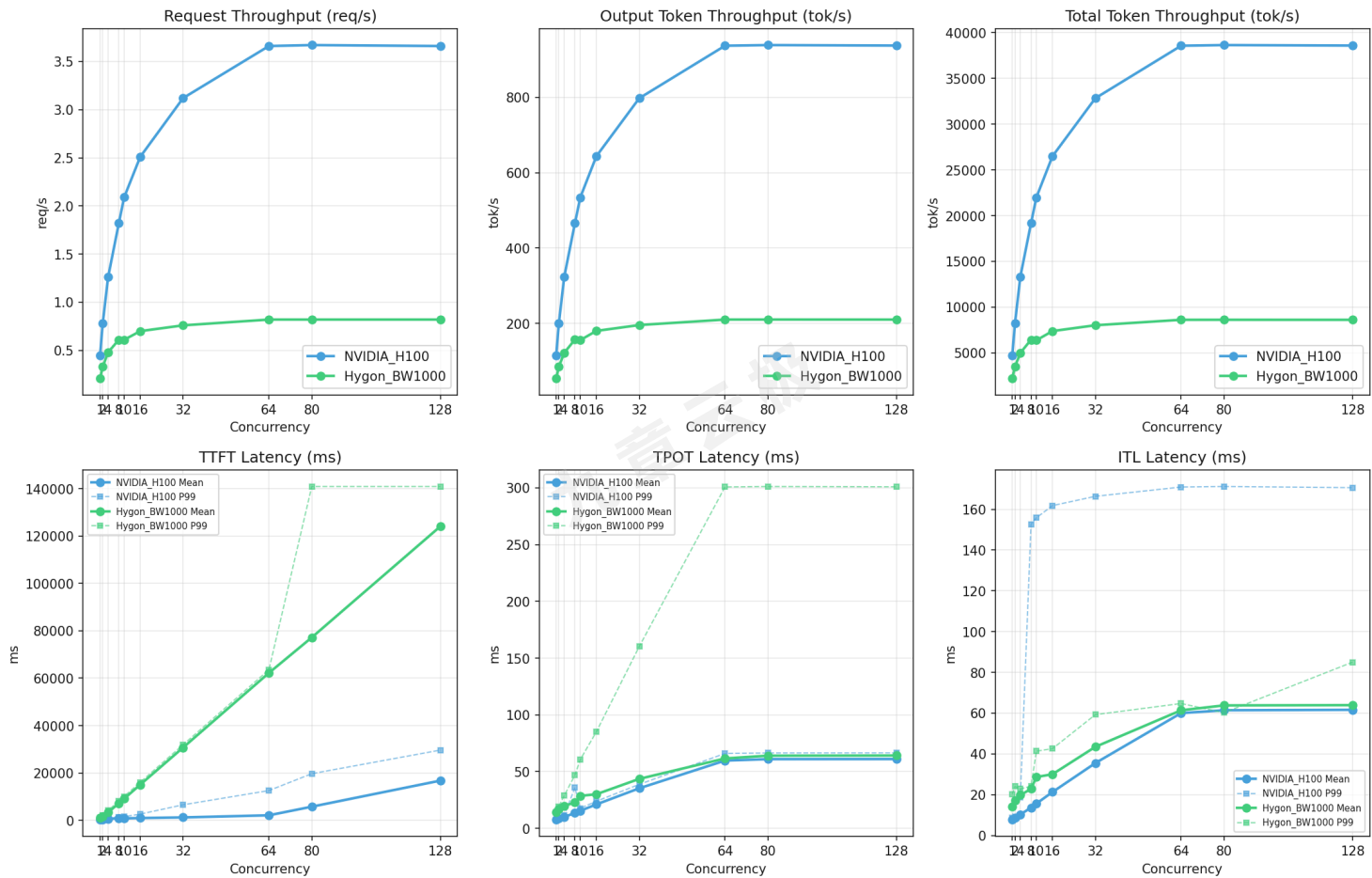
128并发

MiniMax-M2.5 Chip Comparison @ 128 Concurrency



性能趋势对比图 (所有芯片)

MiniMax-M2.5 Performance Trends by Concurrency



各指标随并发级别性能对比详情

请求吞吐量 (Request throughput (req/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	0.45	0.21	-0.24	-53.3%
2	0.78	0.33	-0.45	-57.7%
4	1.26	0.48	-0.78	-61.9%
8	1.82	0.61	-1.21	-66.5%
10	2.09	0.61	-1.48	-70.8%
16	2.51	0.70	-1.81	-72.1%
32	3.12	0.76	-2.36	-75.6%
64	3.66	0.82	-2.84	-77.6%
80	3.67	0.82	-2.85	-77.7%
128	3.66	0.82	-2.84	-77.6%

输出token吞吐量 (Output token throughput (tok/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	115.31	54.28	-61.03	-52.9%
2	199.70	85.05	-114.65	-57.4%
4	323.45	121.84	-201.61	-62.3%
8	465.99	157.18	-308.81	-66.3%
10	534.52	155.66	-378.86	-70.9%
16	643.80	180.19	-463.61	-72.0%
32	797.81	195.69	-602.12	-75.5%
64	937.16	210.25	-726.91	-77.6%
80	938.91	210.32	-728.59	-77.6%
128	937.61	210.20	-727.41	-77.6%

总token吞吐量 (Total token throughput (tok/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	4745.14	2225.40	-2519.74	-53.1%
2	8217.92	3487.19	-4730.73	-57.6%
4	13310.92	4995.56	-8315.36	-62.5%
8	19176.39	6444.25	-12732.14	-66.4%
10	21996.95	6382.12	-15614.83	-71.0%
16	26494.03	7387.79	-19106.24	-72.1%
32	32831.81	8023.25	-24808.56	-75.6%
64	38566.45	8620.30	-29946.15	-77.6%
80	38638.21	8623.30	-30014.91	-77.7%
128	38585.00	8618.28	-29966.72	-77.7%

首token延迟 (P99 TTFT (ms))

九章云极

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	286.01	1168.49	+882.48	+308.5%
2	466.40	2165.32	+1698.92	+364.3%
4	826.89	4137.09	+3310.20	+400.3%
8	1364.44	8133.44	+6769.00	+496.1%
10	1534.23	10081.63	+8547.40	+557.1%
16	2630.84	16063.93	+13433.09	+510.6%
32	6556.86	31917.86	+25361.00	+386.8%
64	12557.76	63531.55	+50973.79	+405.9%
80	19679.20	140904.11	+121224.91	+616.0%
128	29645.32	140903.29	+111257.97	+375.3%

每token生成时间 (P99 TPOT (ms))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	7.68	14.16	+6.48	+84.4%
2	9.05	19.41	+10.36	+114.5%
4	11.41	28.88	+17.47	+153.1%
8	35.95	46.97	+11.02	+30.7%
10	17.75	60.61	+42.86	+241.5%
16	23.92	85.10	+61.18	+255.8%
32	38.79	160.15	+121.36	+312.9%
64	66.03	300.74	+234.71	+355.5%
80	66.45	301.17	+234.72	+353.2%
128	66.52	300.89	+234.37	+352.3%

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	8.57	20.17	+11.60	+135.4%
2	16.54	24.07	+7.53	+45.5%
4	18.61	22.80	+4.19	+22.5%
8	152.68	24.15	-128.53	-84.2%
10	155.84	41.33	-114.51	-73.5%
16	161.83	42.55	-119.28	-73.7%
32	166.38	59.25	-107.13	-64.4%
64	170.95	64.72	-106.23	-62.1%
80	171.22	60.21	-111.01	-64.8%
128	170.62	85.02	-85.60	-50.2%

测试场景二：超长上下文请求测试

测试目标：对超长上下文的请求，使用vllm bench serve工具对并发数逐级增加场景的性能基准验证.

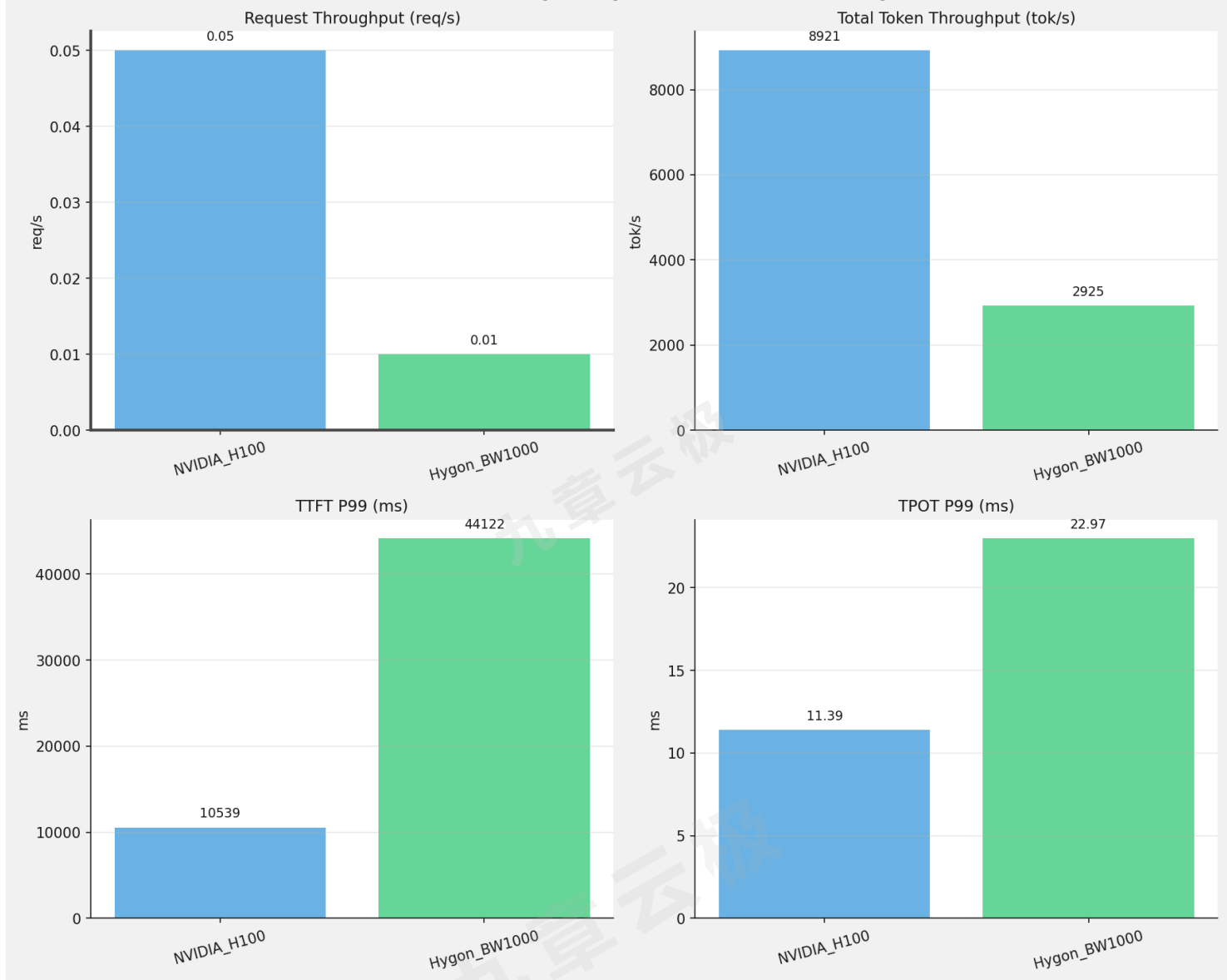
测试概览

项目	配置	备注
数据集	random	
并发数	1, 2, 4, 8, 10	
总请求数	100	
请求输入上下文长度	194560 (190k)	
请求输出上下文长度	1024 (1k)	
被测芯片	NVIDIA_H100, Hygon_BW1000	
被测模型	NVIDIA_H100 (MiniMax-M2.5); Hygon_BW1000 (MiniMax-M2.5-W8A8)	

芯片性能对比柱状图

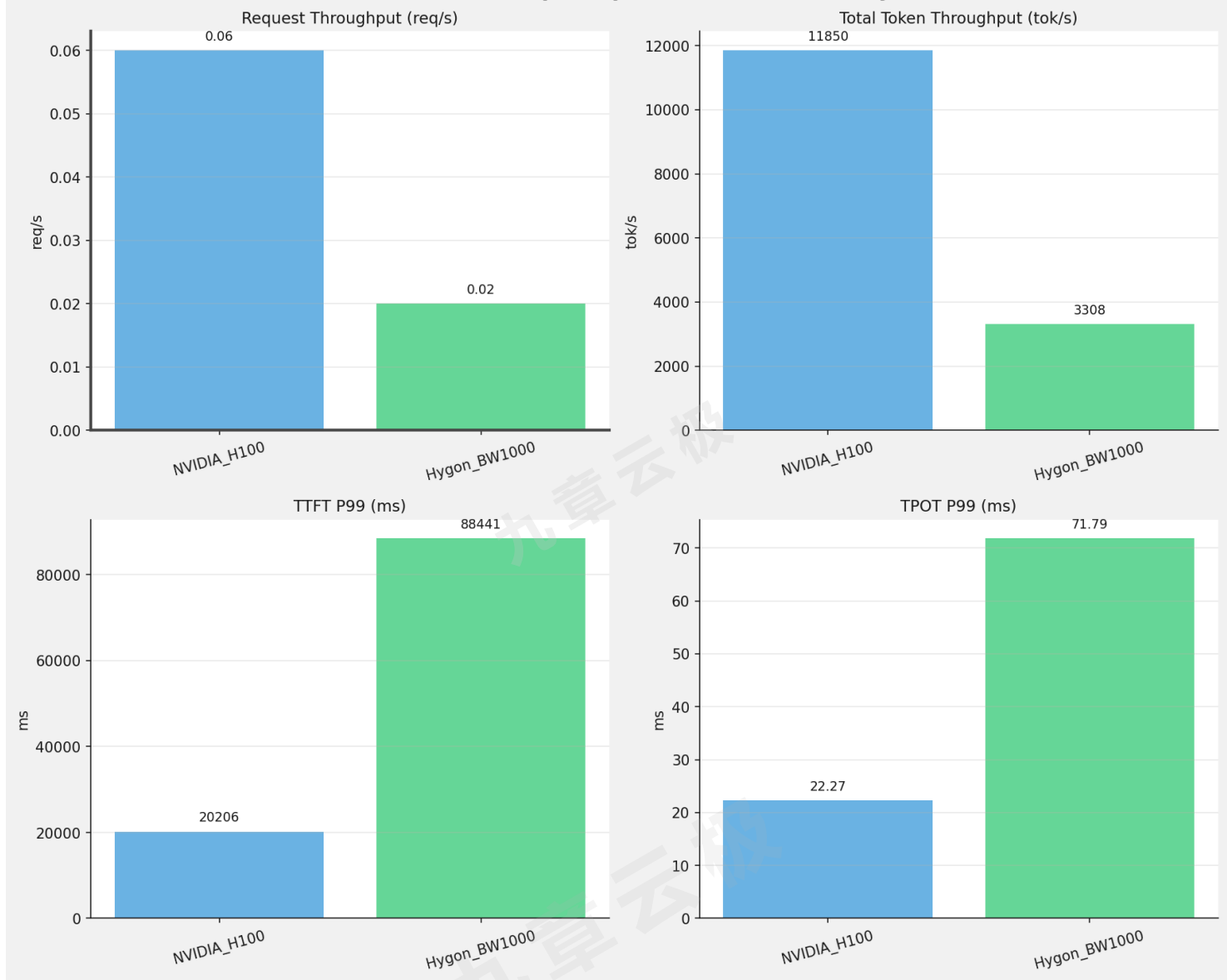
1并发

MiniMax-M2.5 Chip Comparison @ 1 Concurrency



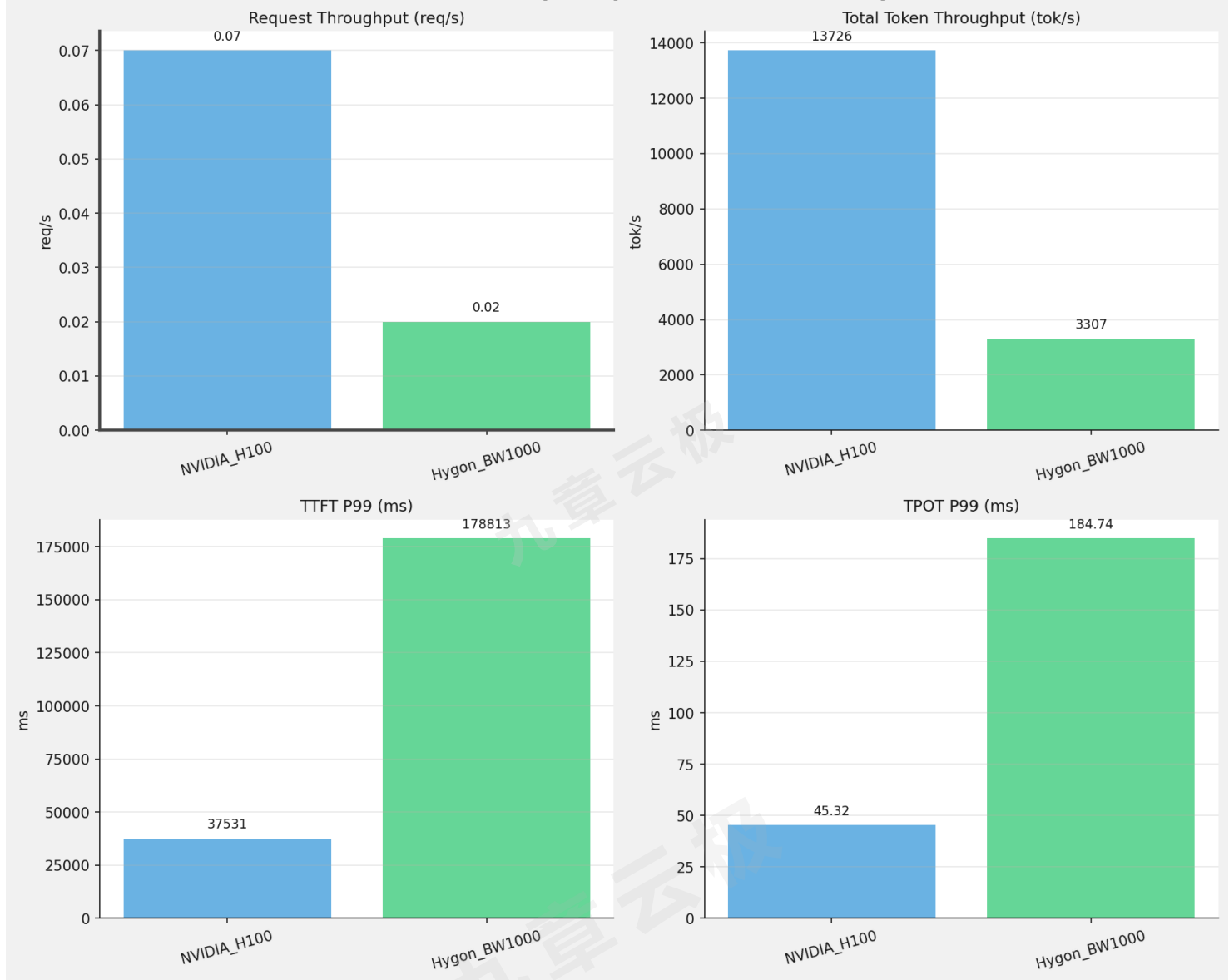
2并发

MiniMax-M2.5 Chip Comparison @ 2 Concurrency



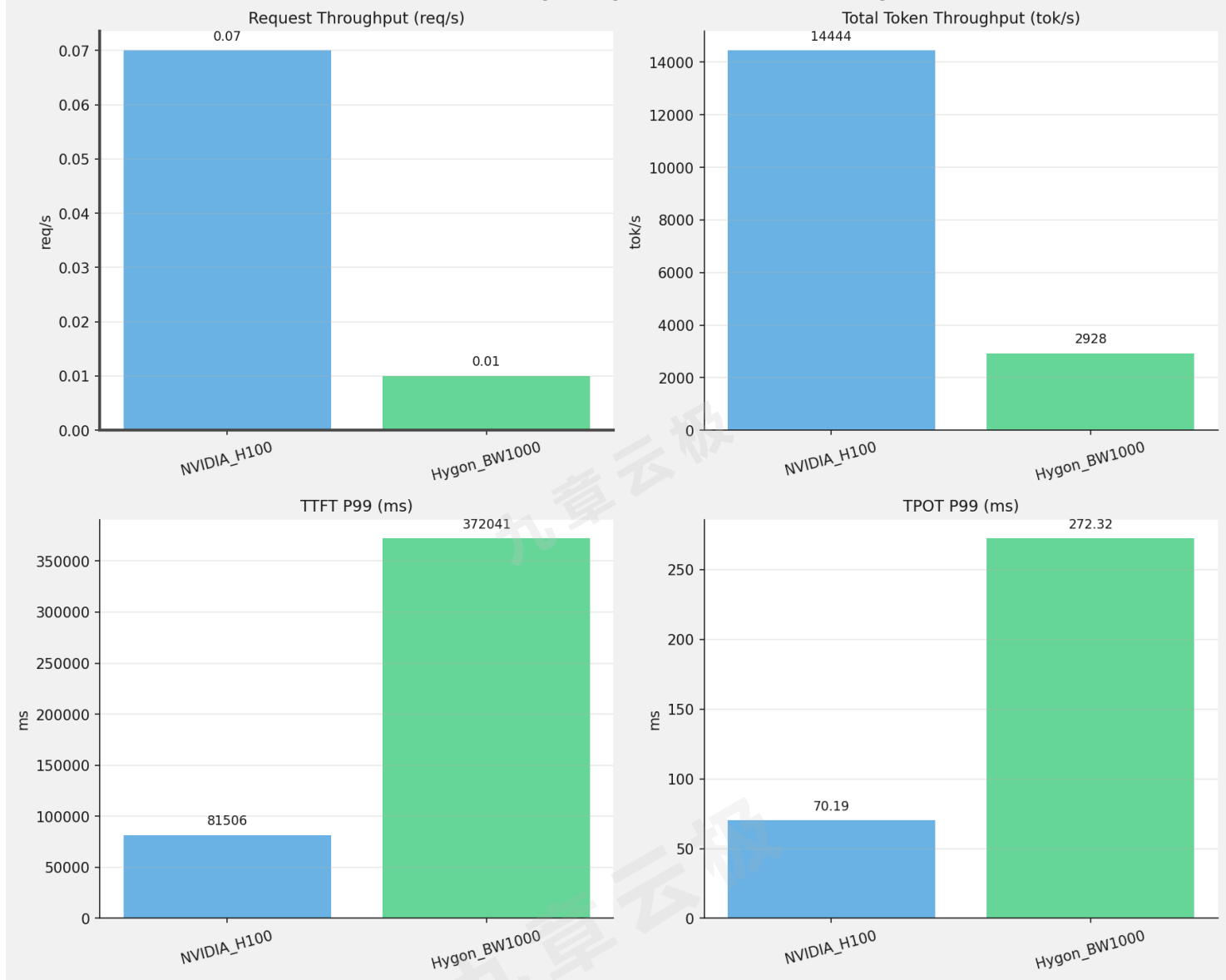
4并发

MiniMax-M2.5 Chip Comparison @ 4 Concurrency



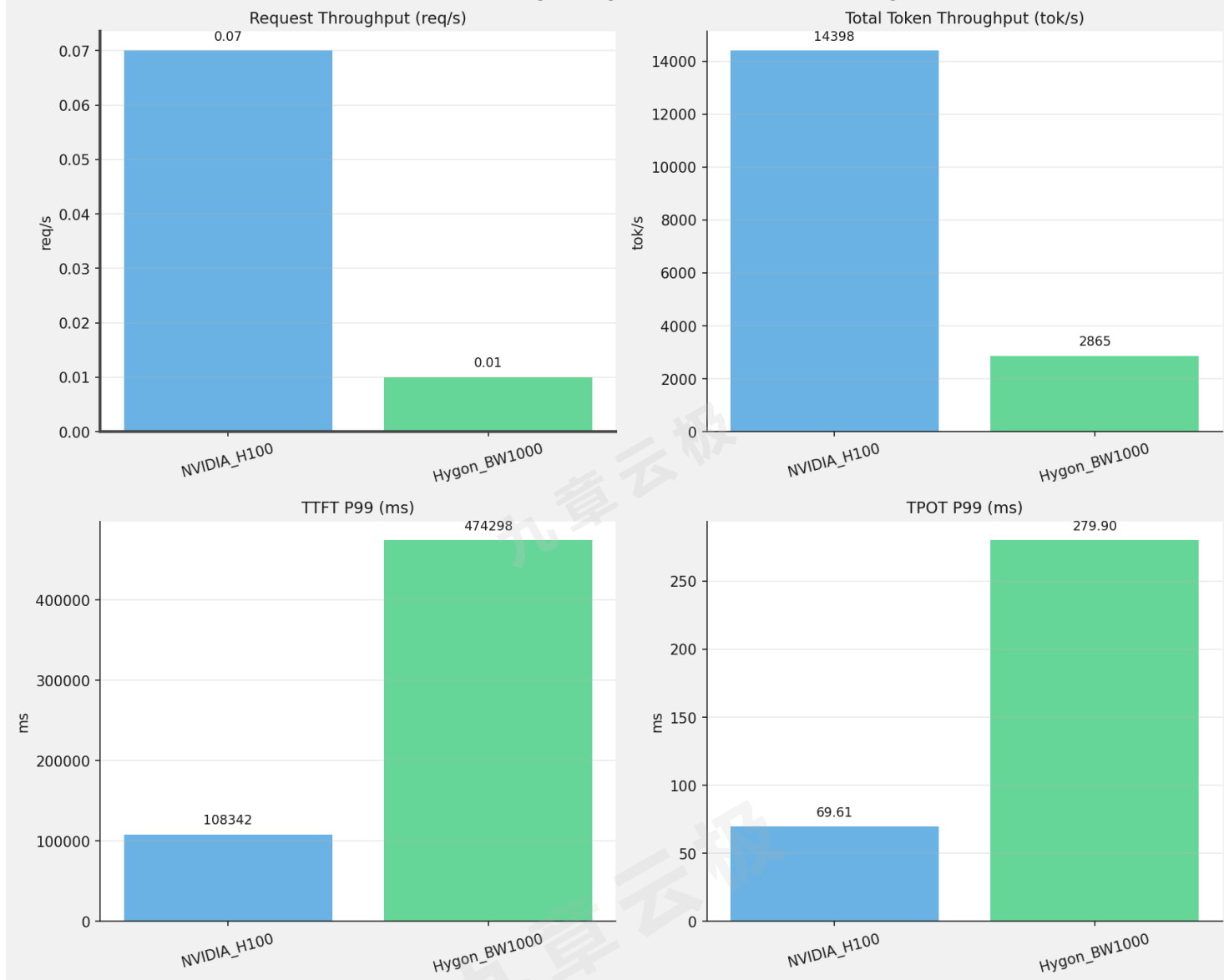
8并发

MiniMax-M2.5 Chip Comparison @ 8 Concurrency



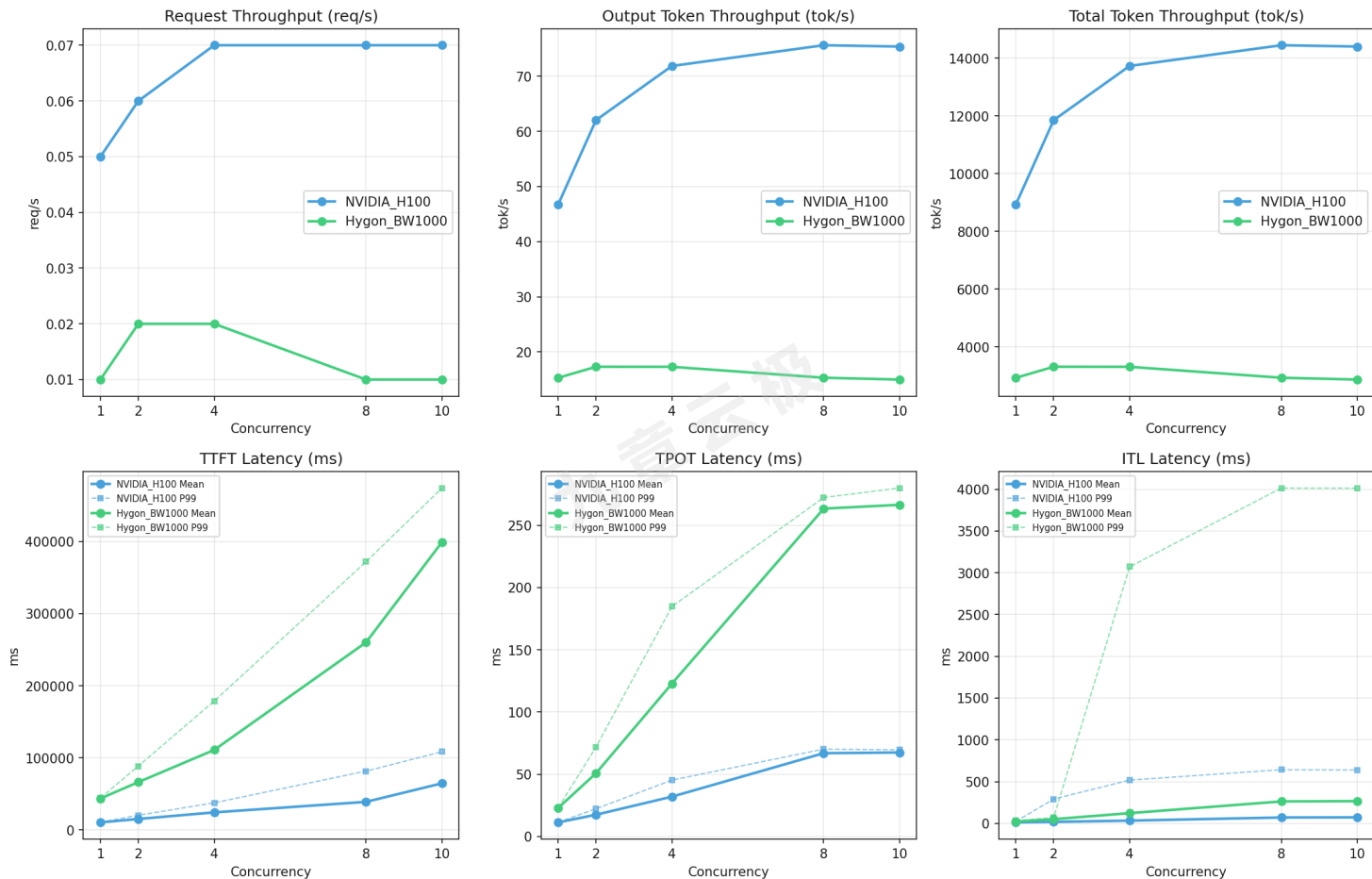
10并发

MiniMax-M2.5 Chip Comparison @ 10 Concurrency



性能趋势对比图 (所有芯片)

MiniMax-M2.5 Performance Trends by Concurrency



各指标随并发级别性能对比详情

请求吞吐量 (Request throughput (req/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	0.05	0.01	-0.04	-80.0%
2	0.06	0.02	-0.04	-66.7%
4	0.07	0.02	-0.05	-71.4%
8	0.07	0.01	-0.06	-85.7%
10	0.07	0.01	-0.06	-85.7%

输出token吞吐量 (Output token throughput (tok/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	46.70	15.31	-31.39	-67.2%
2	62.03	17.32	-44.71	-72.1%
4	71.85	17.31	-54.54	-75.9%
8	75.61	15.33	-60.28	-79.7%
10	75.37	15.00	-60.37	-80.1%

总token吞吐量 (Total token throughput (tok/s))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	8921.40	2924.75	-5996.65	-67.2%
2	11850.06	3308.40	-8541.66	-72.1%
4	13726.45	3306.71	-10419.74	-75.9%
8	14443.68	2927.85	-11515.83	-79.7%
10	14398.41	2865.09	-11533.32	-80.1%

首token延迟 (P99 TTFT (ms))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	10539.10	44121.71	+33582.61	+318.6%
2	20205.87	88441.47	+68235.60	+337.7%
4	37530.99	178812.59	+141281.60	+376.4%
8	81506.35	372041.15	+290534.80	+356.5%
10	108342.29	474297.92	+365955.63	+337.8%

每token生成时间 (P99 TPOT (ms))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	11.39	22.97	+11.58	+101.7%
2	22.27	71.79	+49.52	+222.4%
4	45.32	184.74	+139.42	+307.6%
8	70.19	272.32	+202.13	+288.0%
10	69.61	279.90	+210.29	+302.1%

token间延迟 (P99 ITL (ms))

并发数	NVIDIA_H100	Hygon_BW1000	差值	百分比
1	22.97	32.12	+9.15	+39.8%
2	291.30	69.78	-221.52	-76.0%
4	518.54	3071.54	+2553.00	+492.3%
8	642.87	4014.05	+3371.18	+524.4%
10	639.97	4012.34	+3372.37	+527.0%

测试场景三：长上下文高并发发极限验证

云极

测试目标：多并发长上下文的情况下，验证各芯片单节点同时能处理的最大请求数。

测试概览

项目	配置	备注
数据集	random	
并发数	32, 64	
总请求数	1000	
请求输入上下文长度	90000 (约90k)	
请求输出上下文长度	2000 (约2k)	
被测芯片	NVIDIA_H100, Hygon_BW1000	
被测模型	NVIDIA_H100 (MiniMax-M2.5); Hygon_BW1000 (MiniMax-M2.5-W8A8)	

监控处理请求极限

nvidia_h100芯片

测试平稳后，最大可同时处理请求数13个，且非常稳定，缓存命中率波动正常

(APIServer pid=1007)	INFO 04-30 16:07:48	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 543.4 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.8%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:07:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 544.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.2%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 9000.6 tokens/s, Avg generation throughput: 390.1 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.4%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:10	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 542.1 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.8%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:20	[loggers.py:271]	Engine 000: Avg prompt throughput: 18000.5 tokens/s, Avg generation throughput: 193.5 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 92.0%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:30	[loggers.py:271]	Engine 000: Avg prompt throughput: 26999.2 tokens/s, Avg generation throughput: 42.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 94.4%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:40	[loggers.py:271]	Engine 000: Avg prompt throughput: 27000.0 tokens/s, Avg generation throughput: 42.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 95.9%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:08:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 35803.8 tokens/s, Avg generation throughput: 42.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.1%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 548.5 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.6%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:10	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 547.3 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.0%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:20	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 544.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.5%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:30	[loggers.py:271]	Engine 000: Avg prompt throughput: 9000.7 tokens/s, Avg generation throughput: 390.1 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.6%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:40	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 421.5 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.1%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:09:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 27001.0 tokens/s, Avg generation throughput: 41.6 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.5%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 35999.3 tokens/s, Avg generation throughput: 44.1 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 92.7%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:10	[loggers.py:271]	Engine 000: Avg prompt throughput: 27001.6 tokens/s, Avg generation throughput: 41.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 93.6%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:20	[loggers.py:271]	Engine 000: Avg prompt throughput: 18001.1 tokens/s, Avg generation throughput: 319.4 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.4%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:30	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 547.2 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.8%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:40	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 544.6 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.3%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:10:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 9000.1 tokens/s, Avg generation throughput: 391.4 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.4%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 544.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.9%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:10	[loggers.py:271]	Engine 000: Avg prompt throughput: 17999.6 tokens/s, Avg generation throughput: 148.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 95.5%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:20	[loggers.py:271]	Engine 000: Avg prompt throughput: 27003.8 tokens/s, Avg generation throughput: 41.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.4%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:30	[loggers.py:271]	Engine 000: Avg prompt throughput: 27000.0 tokens/s, Avg generation throughput: 41.6 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 86.7%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:40	[loggers.py:271]	Engine 000: Avg prompt throughput: 35996.8 tokens/s, Avg generation throughput: 91.4 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.2%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:11:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 547.3 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 96.6%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 547.3 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.1%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:10	[loggers.py:271]	Engine 000: Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 531.4 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 91.3%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:20	[loggers.py:271]	Engine 000: Avg prompt throughput: 9000.0 tokens/s, Avg generation throughput: 404.6 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 97.7%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:30	[loggers.py:271]	Engine 000: Avg prompt throughput: 9000.3 tokens/s, Avg generation throughput: 374.3 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 91.6%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:40	[loggers.py:271]	Engine 000: Avg prompt throughput: 27002.1 tokens/s, Avg generation throughput: 42.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 93.2%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:12:50	[loggers.py:271]	Engine 000: Avg prompt throughput: 27001.9 tokens/s, Avg generation throughput: 42.9 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 94.7%, Prefix cache hit rate: 0.0%
(APIServer pid=1007)	INFO 04-30 16:13:00	[loggers.py:271]	Engine 000: Avg prompt throughput: 27002.2 tokens/s, Avg generation throughput: 41.7 tokens/s, Running: 13 reqs, Waiting: 19 reqs, GPU KV cache usage: 95.6%, Prefix cache hit rate: 0.0%

Hygon_BW1000芯片

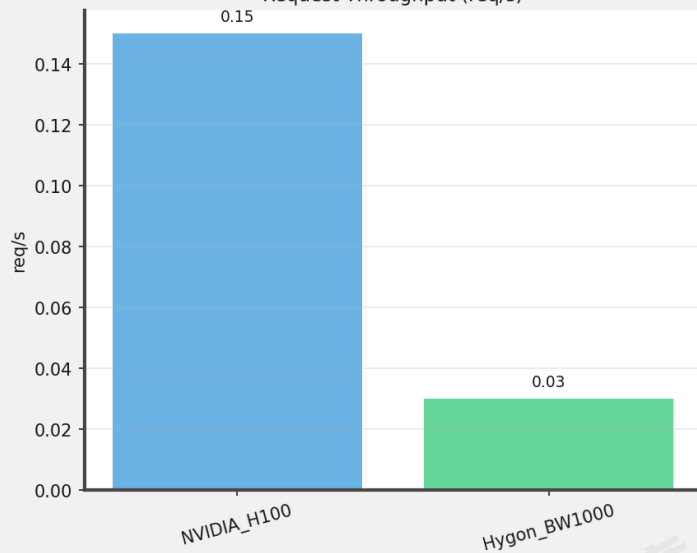
测试平稳后，最大可同时处理请求数在10~18之间波动，缓存命中率正常（在0~3%之间）

[illegible]

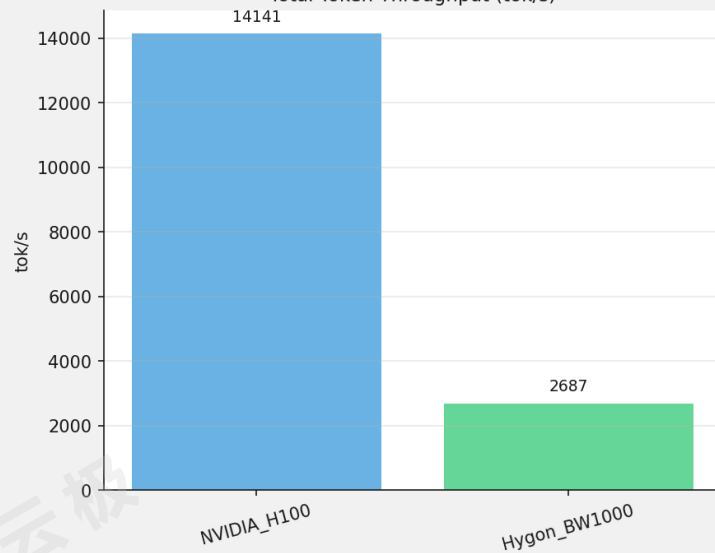
32并发

MiniMax-M2.5 Chip Comparison @ 32 Concurrency

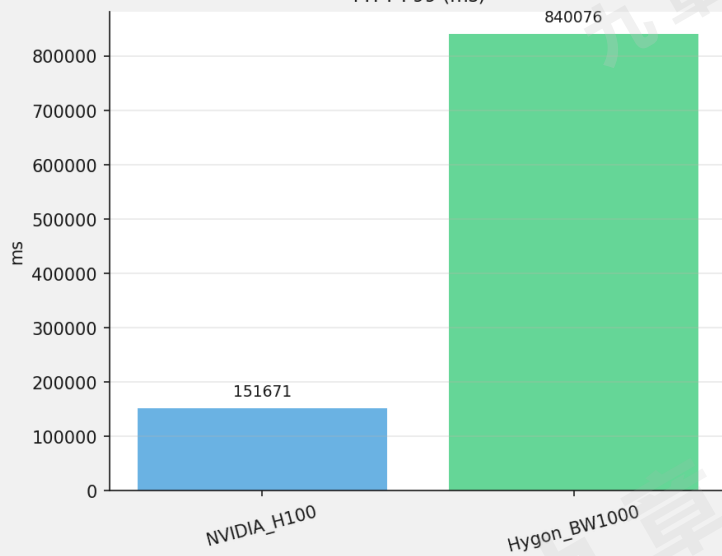
Request Throughput (req/s)



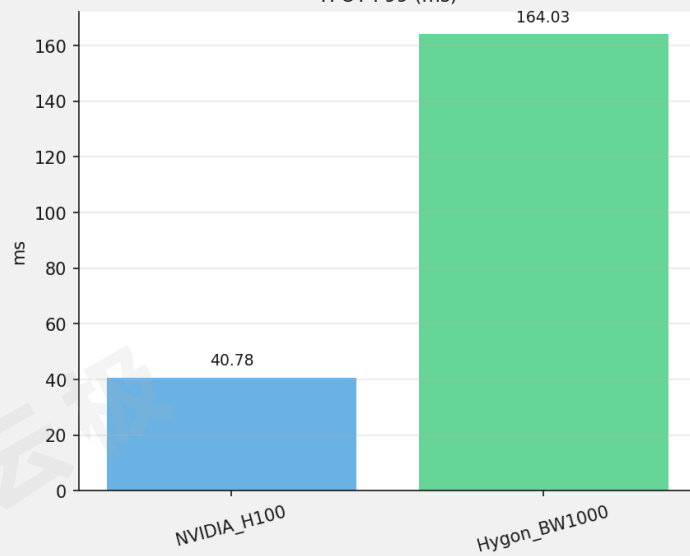
Total Token Throughput (tok/s)



TTFT P99 (ms)

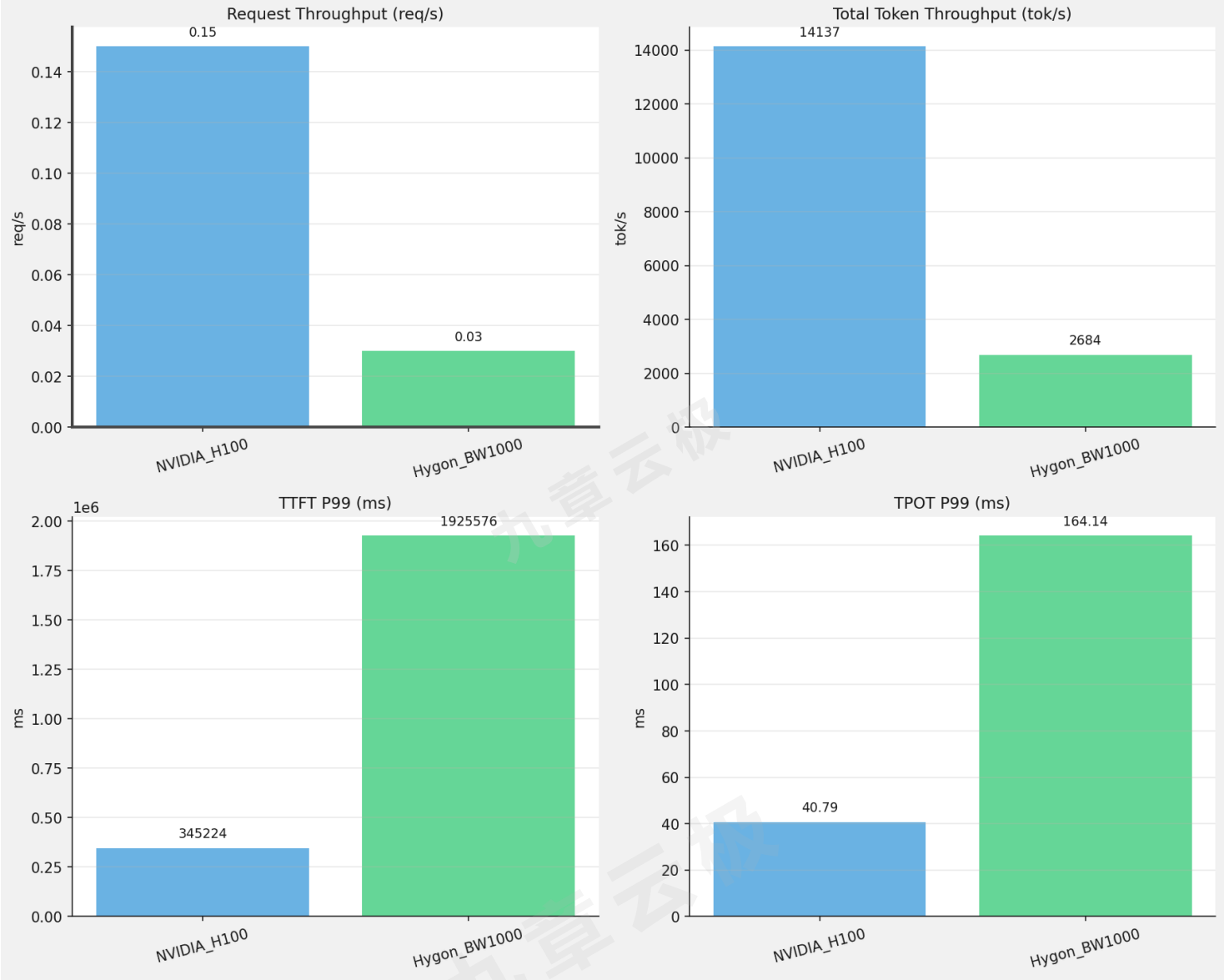


TPOT P99 (ms)



64并发

MiniMax-M2.5 Chip Comparison @ 64 Concurrency



各并发级别性能对比详情

32 并发

指标	NVIDIA_H100	Hygon_BW1000
请求吞吐量（Request throughput (req/s)）	0.15 ⭐	0.03
输出token吞吐量（Output token throughput (tok/s)）	307.27 ⭐	58.42
总token吞吐量（Total token throughput (tok/s)）	14140.63 ⭐	2687.12
首token延迟（P99 TTFT (ms)）	151671.11 ⭐	840076.37
每token生成时间（P99 TPOT (ms)）	40.78 ⭐	164.03
token间延迟（P99 ITL (ms)）	363.93 ⭐	2600.66

指标	NVIDIA_H100	Hygon_BW1000
请求吞吐量 (Request throughput (req/s))	0.15	0.03
输出token吞吐量 (Output token throughput (tok/s))	307.19	58.35
总token吞吐量 (Total token throughput (tok/s))	14136.71	2683.97
首token延迟 (P99 TTFT (ms))	345223.92	1925575.70
每token生成时间 (P99 TPOT (ms))	40.79	164.14
token间延迟 (P99 ITL (ms))	364.12	2600.66

测试场景四：多I/O测试

测试目标

测试不同输入输出长度和并发级别下的性能表现，分析同一芯片同一模型在不同输入输出长度和并发级别下的性能指标变化趋势。

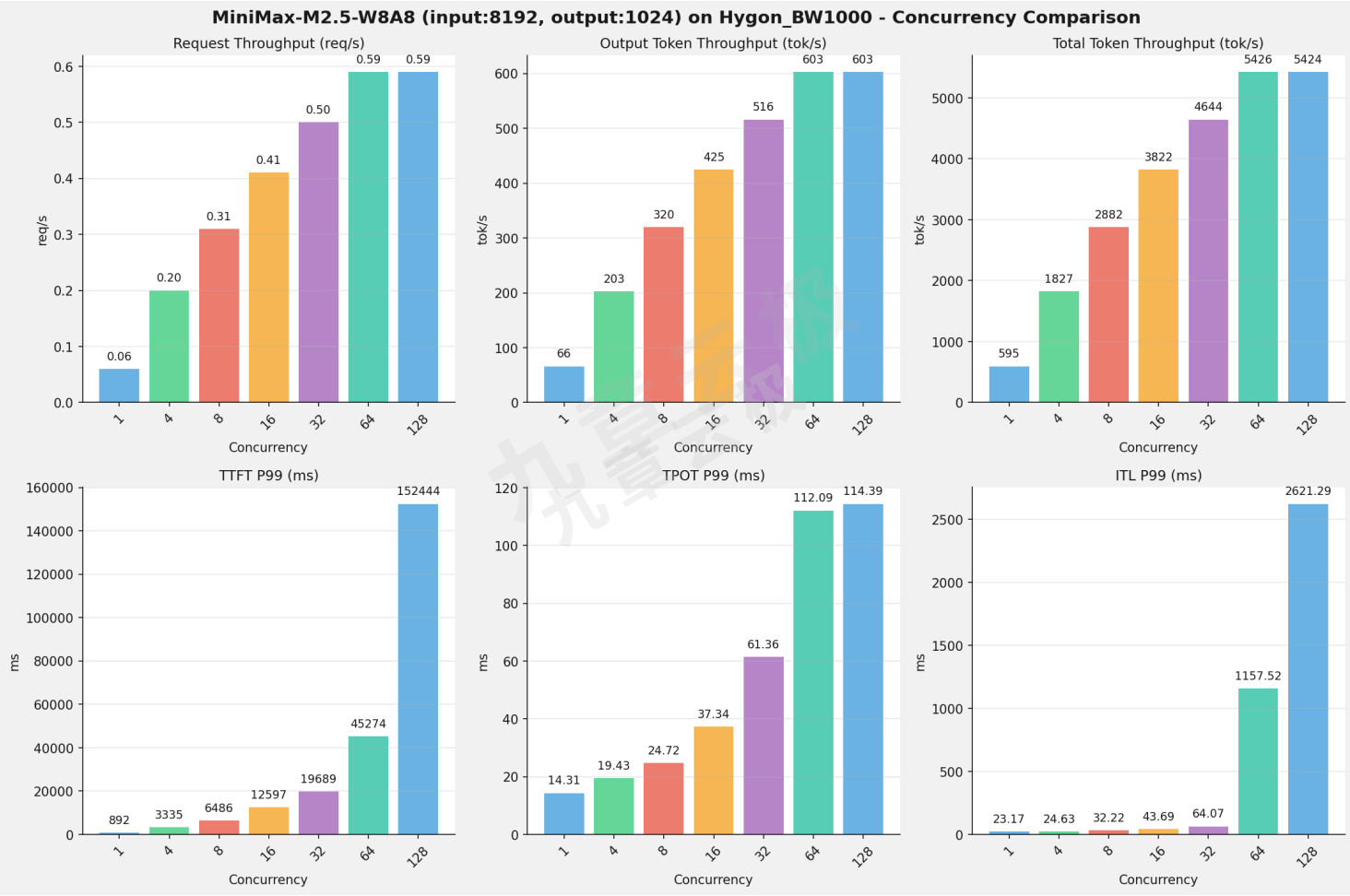
测试概览

项目	配置	备注
数据集	random	
并发数	1, 4, 8, 16, 32, 64, 128	
总请求数	1000	
输入输出长度	(128, 128), (512, 256), (1024, 512), (2048, 1024), (4096, 2048), (8192, 1024)	
被测芯片	NVIDIA_H100, Hygon_BW1000	
被测模型	NVIDIA_H100 (MiniMax-M2.5); Hygon_BW1000 (MiniMax-M2.5-W8A8)	

各I/O测试汇总（固定请求上下文长度，随并发数变化）

报告说明: 由于本测试场景比较多，此报告仅列出一组测试结果作为示例，且仅列出Hygon_BW1000测试结果

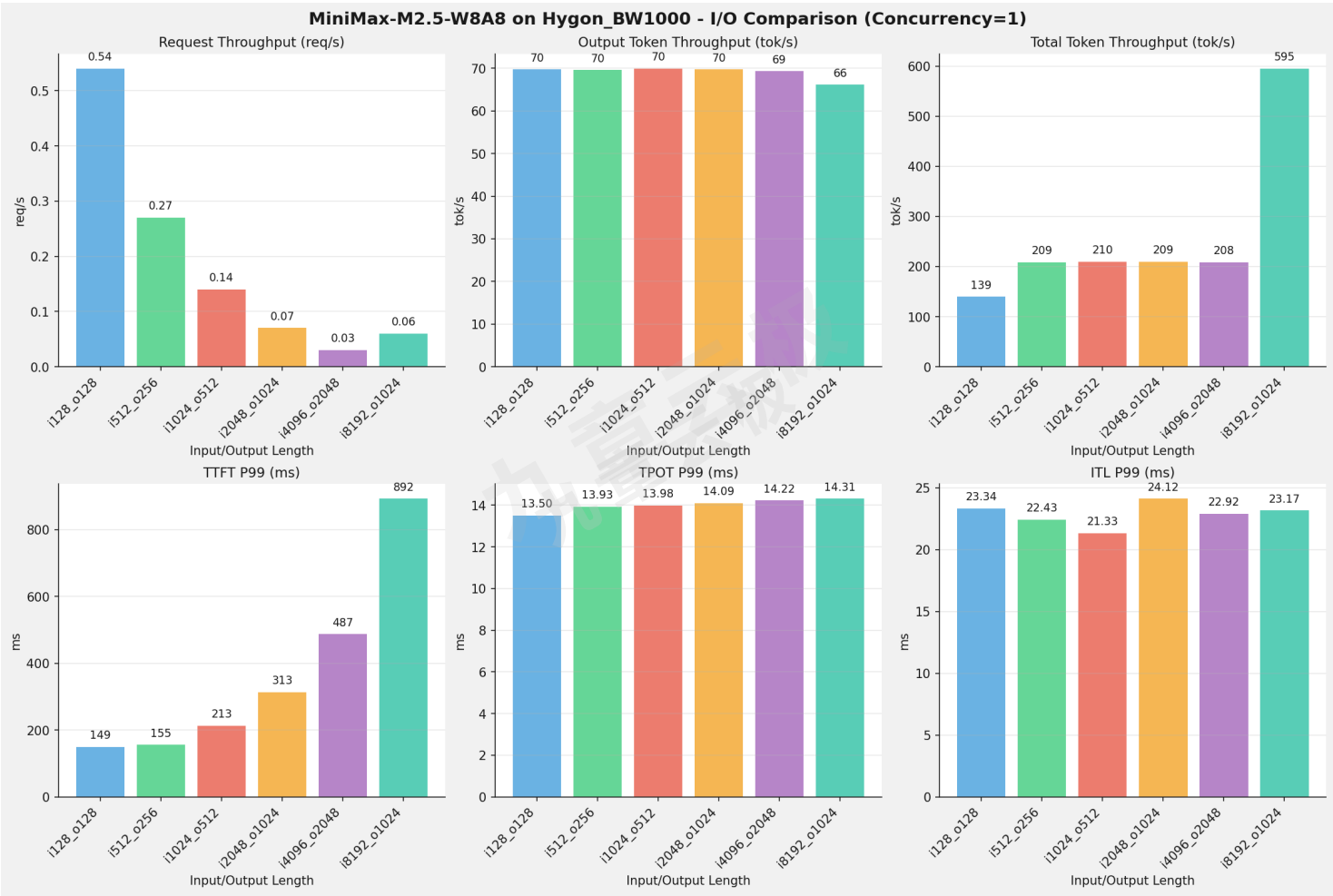
并发数	请求吞吐量 (req/s)	输出Token吞吐量 (tok/s)	总Token吞吐量 (tok/s)	TTFT P99 (ms)	TPOT P99 (ms)	ITL P99 (ms)
1	0.06	66.11	594.97	892.34	14.31	23.17
4	0.20	202.97	1826.69	3335.25	19.43	24.63
8	0.31	320.18	2881.59	6485.86	24.72	32.22
16	0.41	424.67	3822.06	12596.58	37.34	43.69
32	0.50	516.05	4644.47	19688.71	61.36	64.07
64	0.59	602.89	5425.98	45274.45	112.09	1157.52
128	0.59	602.62	5423.60	152443.57	114.39	2621.29



I/O对比（固定并发数, 随请求上下文长度变化）

并发数 = 1

指标	i128_o128	i512_o256	i1024_o512	i2048_o1024	i4096_o2048	i8192_o1024
请求吞吐量 (req/s)	0.54	0.27	0.14	0.07	0.03	0.06
输出Token吞吐量 (tok/s)	69.71	69.57	69.93	69.74	69.33	66.11
总Token吞吐量 (tok/s)	139.42	208.70	209.80	209.22	208.00	594.97
TTFT P99 (ms)	148.98	155.38	213.04	312.85	486.84	892.34
TPOT P99 (ms)	13.50	13.93	13.98	14.09	14.22	14.31
ITL P99 (ms)	23.34	22.43	21.33	24.12	22.92	23.17



测试场景五：模型精度测试

测试目标

模型精度测试目标主要是通过标准指标（如准确率、精确率、召回率、F1值、mAP、AUC）衡量模型在测试集上的输出与真实标签的一致性，评估其基本判别能力。

Task	NVIDIA_H100(FP8)	Hygon_BW1000(W8A8)	差值	百分比
IFBench (Strict)	0.6067	0.2467	-0.3600	- 59.34%
IFBench (Loose)	0.6433	0.2767	-0.3667	- 56.99%
lm-eval:gsm_plus (Flexible)	0.6863	0.6655	-0.0208	- 3.03%
lm-eval:gsm_plus (Strict)	0.7307	0.7301	-0.0006	- 0.08%
lm-eval:mmlu_pro	0.7378	N/A	N/A	N/A
lm-eval:ruler	0.5461	N/A	N/A	N/A

注：海光的测试环境由于无法进行uv虚拟环境依赖安装，mmlu_pro和ruler任务未能执行

- IFBench模型精度测试脚本参见《附录二》
- lm-eval模型精度测试脚本参见《附录三》

测试场景六：多轮对话测试

测试概览

测试工具：vllm 内置的多轮对话测试脚本

条目	值
并发客户端 (num_clients)	50
总对话轮数 (num_conversations)	1000
活跃对话数 (active_conversations)	100
输入轮数 (input_num_turns)	10
输入 Token 长度 (input_num_tokens)	200
输出 Token 长度 (output_num_tokens)	100
语料来源	推荐的语料 pg1184.txt
seed	0

和英伟达比对

指标	NVIDIA_H100	Hygon_BW1000
总耗时	140.633 秒	416.808 秒
请求吞吐量 （RPS）	28.087	9.477

指标	count	mean	std	99%	99.9%	max
ttft_ms	3950	1334.39	489.78	2497.44	2869.67	5106.10
tpot_ms	3950	38.24	6.24	63.09	69.61	80.53
latency_ms	3950	5265.98	550.01	8109.88	8233.94	9837.93
input_num_turns	3950	5.96	2.22	9.00	9.00	9.00
input_num_tokens	3950	949.43	336.67	1419.00	1425.00	1430.00
output_num_tokens	3950	103.91	2.07	110.0	112.05	120.00
output_num_chunks	3950	97.99	5.28	99.00	99.00	99.00

多轮对话测试脚本参见本报告《附录四》

测试场景七：基础推理能力验证

测试目标：

验证模型在芯片环境上的基础推理和兼容性的支持能力情况，可作为快速选型的一个基础指标。

测试说明

比对说明：本章节只列出在Hygon_BW1000芯片平台上的测试结果

状态说明：✅ 已通过，⌚ 未测试，❌ 未通过，⚠️ 部分通过

A. 基础推理能力

#	测试点	测试内容	状态	备注
A1	单轮对话	发送单条prompt，验证正常生成	✅	
A2	多轮对话	5轮对话，验证上下文保持和连贯性	✅	
A3	System Prompt	设置系统角色，验证模型遵循程度	✅	
A4	流式输出	stream=true，验证SSE逐token返回	✅	
A5	非流式输出	stream=false，验证完整返回	✅	
A6	Temperature 控制	temp=0 vs temp=1.0，验证输出差异	✅	
A7	Top-p/Top-k采样	不同top_p/top_k值，验证多样性控制	✅	
A8	Max Tokens限制	设置max_tokens，验证输出不超限	✅	
A9	Stop Sequences	设置stop token，验证截断	✅	
A10	Seed 可复现性	相同seed+temp=0，验证输出一致	✅	
A11	多语言能力	中/英/日/韩/法等多语言输入输出	✅	
A12	特殊Token处理	含emoji、代码块、数学符号、HTML标签	✅	

B. 高级生成功能

#	测试点	测试内容	状态	备注
B1	思考模式 (Thinking)	开启thinking mode, 验证返回思考链...	✓	
B2	非思考模式 (Instant)	关闭thinking, 验证无hidden th...	✗	关闭思考模式，依然有思考内容以<\think>结尾出现在content里
B3	思考模式切换	同一会话内thinking↔non-think...	✗	使用默认temperature 0.7
B4	工具调用-单工具	定义单个function, 验证模型正确调用并传参	✓	
B5	工具调用-多工具	定义多个function, 验证模型选择正确的工具	✗	未正常调用多个工具
B6	工具调用-并行调用	单次回复中并行调用多个工具	✓	
B7	工具调用-多步链式	工具结果作为下一步输入, 验证3+步链式执行	✗	未正常进行链式工具调用
B8	JSON Mode	response_format=json_ob...	✓	
B9	结构化输出	JSON Schema约束输出格式, 验证字段完整性	✓	
B10	Prefix/Suffix约束	指定输出前缀或格式模板, 验证遵循度	✗	

C. 多模态能力（MiniMax-M2.5模型为文本模型，此测试组请跳过）

#	测试点	测试内容	状态	备注
C1	单图理解	图片+文本提问	✗	
C2	多图对比	跨图比较	✗	
C3	高分辨率图片	4K分辨率	✗	
C4	图表/OCR	表格截图	✗	
C5	视频理解	视频文件	✗	
C6	代码截图→代码	UI截图	✗	
C7	多模态工具调用	图片触发工具	✗	
C8	图片格式兼容性	PNG/JPEG/WebP	✗	

#	测试点	测试内容	状态	备注
D1	短上下文基线	1K tokens	✓	
D2	中等上下文	8K-16K tokens	✓	
D3	长上下文	32K-64K tokens	✓	
D4	超长上下文	128K+ tokens	✓	
D5	大海捞针	NIAH	✓	
D6	上下文边界行为	max_model_len	✓	
D7	超出上下文截断	截断/拒绝	✓	
D8	长输出生成	4K-8K tokens	✓	

F. 稳定性与边界

#	测试点	测试内容	状态	备注
F1	空输入	空prompt	✓	返回 Empty message handled: We have a conversation. The user wrote only " "
F2	超大输入	超max_model_len	✓	返回 Oversized input handled, finish_reason: length
F3	非法参数	temperature=-1, max_tokens=0,非法温度值：超过范围（>2）	✓	返回 API request failed with status 400
F4	特殊字符注入	SQL/Prompt注入	✓	
F5	并发稳定性	200+并发（实际测试50并发）	✓	
F6	OOM恢复	显存耗尽	⌚	
F7	长时间运行	24小时	⌚	
F8	请求超时处理	超时断开	⌚	

#	测试点	测试内容	状态	备注
G1	OpenAI Chat Completions	/v1/chat/completions 接口兼容	✓	
G2	OpenAI Completions	/v1/completions 接口兼容	✓	
G3	模型列表	/v1/models 返回可用模型	✓	
G4	Usage 统计	usage 字段准确	✓	
G5	错误码规范	400/401/404/429/500 错误码	⌚	
G6	客户端 SDK 兼容	Python openai / JS @ope...	⌚	
G7	响应格式变体	不同response_format	✓	
G8	Stream参数	stream参数测试	✓	

H. 质量评估

#	测试点	测试内容	状态	备注
H1	生成质量	质量对比	✓	
H2	生成一致性	多次生成	✓	
H3	幻觉率	事实错误	✓	
H4	指令遵循度	格式/角色	✓	
H5	响应相关性	问答相关性	✓	

I. 超长上下文验证

#	测试点	测试内容	状态	备注
I1	超长上下文（非流式）	验证超长上下文请求的非流式输出	✓	
I2	超长上下文（流式）	验证超长上下文请求的流式输出	✓	
I3	超长上下文（边界验证）	使用二分法逼近模型最大上下文长度	✓	
I4	超长上下文（思考模式）	验证超长上下文下reasoning_conte...	✓	

附录一： benchmark执行脚本

执行命令

```
python run_benchmark.py --chip hygon_bw1000 --model MiniMax-M2.5-w8A8 --test-suite test_01,test_02,test_03,test_04,test_05

# 如果不指定test-suite参数，默认执行run_benchmark.py里TEST_SUITES定义的测试套件列表，可以根据自己需要修改默认测试套件
```

```
vllm bench serve
--backend openai-chat
--endpoint /v1/chat/completions
--dataset-name random
--random-input-len 10240
--random-output-len 256
--model ${MODEL_PATH}
--trust-remote-code
--base-url ${BASE_URL}
--num-prompts 320
--max-concurrency 1
--temperature 0.7
--seed 123
--metric_percentiles 95,99
--served-model-name ${MODEL}
--ready-check-timeout-sec 30
```

附录二：IFBench精度测试脚本

以MiniMax-M2.5-W8A8模型为例

ifbench_mm25_w8a8.sh

```
#!/bin/bash
ROOT_PATH=$(cd `dirname $0`; pwd)

echo $ROOT_PATH
cd ${ROOT_PATH}

CurDate=`date +%Y%m%d`
export NLTK_DATA=${ROOT_PATH}/nltk_data

cat > .env << 'EOF'
api_base=http://127.0.0.1:8000/v1
api_key=abc123
model=minimax-m2.5
temperature=1.0
top_p=0.95
top_k=40
max_tokens=8192
seed=42
input_file=data/IFBench_test.jsonl
output_file=data/mm25-responses.jsonl
workers=32
EOF

# 2. 生成模型响应
uv run python generate_responses.py

# 3. Thinking 模型后处理（重要！）
uv run python postprocess_thinking.py data/mm25-responses.jsonl -o data/mm25-clean.jsonl

# 4. 运行评估
uv run python -m run_eval \
    --input_data=data/IFBench_test.jsonl \
    --input_response_data=data/mm25-clean.jsonl \
    --output_dir=eval
```

以MiniMax-M2.5-W8A8模型为例

Im_eval_test.sh

```
#!/bin/bash
ROOT_PATH=$(cd `dirname $0`; pwd)

echo $ROOT_PATH
cd ${ROOT_PATH}

CurDate=`date +%Y%m%d`

export HF_HUB_OFFLINE=1
export TRANSFORMERS_OFFLINE=1

#export HF_ENDPOINT=https://hf-mirror.com

ADDR=${ADDR:-127.0.0.1}
PORT=${PORT:-8000}
API_KEY=${API_KEY:-abc123}
LLM_ADDR="http://$ADDR:$PORT"

# 自动获取模型名和 tokenizer 路径
#MODEL_NAME=$(curl -s --header "Authorization: Bearer $API_KEY" $LLM_ADDR/v1/models | jq -r .data[0].id)
#MODEL_PATH=$(curl -s --header "Authorization: Bearer $API_KEY" $LLM_ADDR/v1/models | jq -r .data[0].root)
MODEL_NAME="minimax-m2.5"
LOCAL_MODEL_PATH="/data/models/MiniMax-M2.5-W8A8"

# model_args 构造
MODEL_ARGS_BASE_1="
{\"model\": \"$MODEL_NAME\", \"base_url\": \"$LLM_ADDR/v1/completions\", \"max_length\": 131072, \"tokenizer\": \"$LOCAL_MODEL_PATH\", \"trust_remote_code\": true, \"num_concurrent\": 10, \"max_retries\": 3, \"timeout\": 12000, \"tokenized_requests\": false, \"headers\": {\"Authorization\": \"Bearer $API_KEY\"}}"
MODEL_ARGS_BASE_2="
{\"model\": \"$MODEL_NAME\", \"base_url\": \"$LLM_ADDR/v1/completions\", \"max_length\": 192512, \"tokenizer\": \"$LOCAL_MODEL_PATH\", \"trust_remote_code\": true, \"num_concurrent\": 10, \"max_retries\": 3, \"timeout\": 12000, \"tokenized_requests\": false, \"headers\": {\"Authorization\": \"Bearer $API_KEY\"}}"

# 运行单个任务的函数
run_task_1() {
    local task_name=$1
    local max_tokens=$2
    local temperature=$3
    local unsafe_code=$4

    local do_sample="false"
    [ "$temperature" = "1.0" ] && do_sample="true"

    GEN_KWARGS="
{\"max_gen_toks\": $max_tokens, \"do_sample\": $do_sample, \"temperature\": $temperature, \"top_p\": 0.95, \"top_k\": 40}"

    local unsafe_flag=""
    [ "$unsafe_code" = "true" ] && unsafe_flag="--confirm_run_unsafe_code" && export HF_ALLOW_CODE_EVAL=1

    lm_eval \
        --model local-completions \
        --tasks $task_name \
        --output_path ./output/${task_name}/${MODEL_NAME}_${CurDate} \
```

```

--model_args "$MODEL_ARGS_BASE_1" \
--batch_size auto \
--gen_kwargs "$GEN_KWARGS" \
$unsafe_flag
}

run_task_2() {
    local task_name=$1
    local max_tokens=$2
    local temperature=$3
    local unsafe_code=$4

    local do_sample="false"
    [ "$temperature" = "1.0" ] && do_sample="true"

    GEN_KWARGS="
{\"max_gen_toks\":$max_tokens,\"do_sample\":$do_sample,\"temperature\":$temperature,\"top_p\":0.95,\"top_k
\":40}"

    local unsafe_flag=""
    [ "$unsafe_code" = "true" ] && unsafe_flag="--confirm_run_unsafe_code" && export HF_ALLOW_CODE_EVAL=1

    lm_eval \
        --model local-completions \
        --tasks $task_name \
        --output_path ./output/${task_name}/${MODEL_NAME}_${CurDate} \
        --model_args "$MODEL_ARGS_BASE_2" \
        --batch_size auto \
        --limit 32 \
        --gen_kwargs "$GEN_KWARGS" \
        $unsafe_flag
}

run_task_1 mmlu_pro 8192 0.0 false

sleep 120
run_task_1 gsm_plus 8192 0.0 false

sleep 120
run_task_2 ruler 8192 0.0 false

```

附录四：多轮对话测试命令

```

python benchmark_serving_multi_turn.py \
--model /data/models/MiniMax-M2.5-w8A8 \
--served-model-name minimax-m2.5 \
--url http://127.0.0.1:8080/v1 \
--input-file generate_conversations.json \
--output-file generate_conversations_output.json \
--num-clients 50 \
--max-active-conversations 100 \
--warmup-step \
--extra-body-json {"chat_template_kwargs": {"enable_thinking": false}}

```